

Enhancing differential evolution algorithm for CEC 2014, CEC 2017, CEC 2021, and CEC 2022 test suites

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Abstract

Differential evolution (DE) has demonstrated its significant contribution to the optimization of different real-world applications as well as standard benchmarks. This paper presents a novel variant of the DE algorithm, known as the LSHADESPA algorithm. The LSHADESPA algorithm incorporates three significant modifications to enhance its performance. Firstly, a proportional shrinking population mechanism is employed to reduce the computational burden. Secondly, a simulated annealing (SA)-based scaling factor is introduced to improve the exploration properties of the algorithm. Finally, an oscillating inertia weight-based crossover rate is utilized to strike a balance between exploitation and exploration. These modifications aim to enhance the overall efficiency and effectiveness of the DE algorithm. The proposed LSHADESPA algorithm has been empirically evaluated on a set of benchmark problems, namely the CEC 2014, CEC 2017, CEC 2021, as well as CEC 2022. The experimental outcomes show that the LSHADESPA algorithm performs superior to other MH algorithms. Additionally, the Wilcoxon rank-sum as well as the Friedman rank test proves the statistical significance of the proposed LSHADESPA in comparison to other algorithms under comparison. The outcomes indicate that the LSHADESPA algorithm has statistical significance, with Friedman statistics for the CEC 2014, CEC 2017, and CEC 2022 benchmark functions achieving the lowest f-rank value compared to the other MH algorithms which are found to be 41, 77, and 26, respectively, and obtained 1st rank. Note that this paper is an invited extended version of the paper published in ISCFI 2022 conference.

Keywords Evolutionary algorithm (EA) · Differential evolution (DE) · LSHADE · CEC benchmark problems · Optimization

Abbreviations

DE	Differential evolution
IW	Inertia weight
SP	Shrinking population
MH	Meta-heuristic
SA	Simulated annealing
CR	Crossover rat
LP	Linear programming
QP	Quadratic programming
NLP	Nonlinear programming
IP	Integer programming
EA	Evolutionary algorithm



GA	Genetic algorithms
OF	Objective function
GP	Genetic programming
EDA	Estimation of distribution algorithm
D	Dimension
IMEHO	Improved elephant herding optimization
NP	Population size
MVMO	Mean-variance mapping optimization
$t_{\max}$	Maximum iterations
RW-GWO	Random walk gray wolf optimizer
NFL	No free lunch theorem
PBIL	Population-based incremental learning
SaDE	Self-adaptive DE
B-BBO	Blended biogeography-based optimization
std	Standard deviation
ISOS	Improved symbiotic organisms search
PSO	Particle swarm optimization
KOA	Kepler optimization algorithm
LSO	Light spectrum optimizer
VNBA	Variable neighborhood bat algorithm
ES	Evolution strategies
CEC	Congress on evolutionary computation
CCS	Chaotic cuckoo search

1 Introduction

Optimization [1] is a mathematical discipline focused on finding the best solution to a problem out of all feasible solutions. It involves maximizing desired factors and minimizing undesired ones to reach the optimal design, decision, or operation. Optimization has applications across engineering, economics, operations research, control theory, and other quantitative fields. The key components of an optimization [2] problem are: Objective function (*OF*) is the quantity to be maximized or minimized, and it represents the goal or performance metric, decision variables are those variables that can be set or changed to affect the *OF* value, and finally, constraints denote limits or conditions the variables must satisfy. The optimization statement combines these elements: Minimize/Maximize $f(x)$ Subject to constraints by choosing x . There are various classifications of optimization problems that have specific algorithms developed to solve them efficiently: (a) Linear programming (LP) (*OF* and constraints are linear functions) [3], (b) Quadratic programming (QP) (*OF* is a quadratic function [4], constraints are linear), (c) nonlinear programming (NLP) (*OF* or constraints are nonlinear functions [5]), (d) integer programming (IP) (decision variables must take integer values) [6], and (e) stochastic programming (involves uncertainty modeled using random variables) [7]. Some common mathematical techniques used to solve optimization problems include: (a) calculus methods, (b) Direct search, (c) Convex optimization, (d) Dynamic programming, and (e) Heuristics [8].

Optimization algorithms can be broadly categorized into two main classes: derivative-based as well as derivative-free methods. Derivative-based methods like gradient descent are faster but require the *OF* to be differentiable. Derivative-free methods like evolutionary algorithms are more flexible. Key trends involve gradient-free, stochastic, adaptive, hybrid, meta-learning, and quantum techniques [9–11]. These expand capabilities

and improve efficiency and robustness across diverse challenges. To identify optimal or near-optimal solutions to complicated problems, a meta-heuristic (MH) search algorithm replicates the stages of natural evolution. There are two types of MH search algorithms: local as well as global search. Local search techniques begin with an initial solution and enhance it repeatedly by investigating its neighborhood, i.e., a subset of the search space which is near to the present solution. And global search algorithms retain a population of solutions and employ genetic operators such as crossover as well as mutation to generate new solutions from current ones. The right balance of exploration as well as exploitation searches is the foundation for all MH algorithms to properly tackle various sorts of optimization challenges. Existing MH algorithm may be categorized into four primary types: (a) evolutionary algorithm (EAs), (b) swarm intelligence computational methods, (c) human-oriented algorithms as well as (d) physics-oriented algorithms [12, 13]. Evolution of EAs is driven by Darwin's theory called "survival of the fittest." These EAs use crossover, mutation, and selection to produce new offspring solutions of higher quality. Evolutionary algorithms [14] are a family of population-based optimization algorithms inspired by biological evolution. The key characteristics EAs are population of candidate solutions, selection based on fitness, variation through recombination as well as mutation, and iterative improvement across generations. The major classes of EAs are: Genetic algorithms (GA) [15], Evolution strategies (ES) [16], Genetic programming (GP) [17], Differential evolution (DE), Estimation of distribution algorithms (EDA) [18]. The present paper proposes a self-adaptive DE algorithm. DE is a population-based search algorithm that enhances candidate solutions based on evolutionary principles like mutation, crossover, as well as selection. The various review study related to the DE algorithm is discussed as follows. Neri et al. [19] were among the first to review the research progress on DE, focusing on a study of several enhanced DE structures with extra components released up to 2009, in addition to their ability in solving selected conventional as well as rotating problems of diverse dimensional sizes. Das et al. [20, 21] released two review articles to carefully examine the advances in DE research up to 2011–2016, correspondingly. The key themes discussed in these review articles are adjustments made to current DE variations to tackle various sorts of optimization environments. Also, the review paper released by Jebaraj et al. [22] focuses on the potential uses of DE to tackle static as well as dynamic economic or emission dispatch challenges reported up to 2016. The issue articulation in perspective of the *OF*, in addition to the equality as well as inequality restrictions associated to various economic alongside emission dispatch concerns, was first presented, followed by a description of current DE variants aimed to address these problems. The review study by Opara et al. [23] concentrated on assessments of DE, compared to most current research that examined alterations performed on DE variations along with technical aspects. This study fully covered substantial quantities like convergence characteristics, computational complexity, population diversity, as well as population dynamics frameworks related to DE published till 2019. In the same year, Javaid [24] released a review study; however, it only included DE versions along with their uses in energy control challenges till 2016. Review by Bilal et al. [25] highlighted the most recent advances in DE investigations reported till 2018. Aside from the adjustments made to the initial DE, this attempt performed a full bibliometric assessment of the DE in order to provide its publishing numbers based on various journal quartiles as well as publishers.

The main DE variants are as follows. JADE uses self-adaptive control to adjust control parameters F and CR [26]. SHADE introduces history-based adaptive parameter control and external archive [27]. LSHADE incorporates linear population size reduction to improve convergence. LSHADE-EpSin [28] uses ensemble sinusoidal parameter adaptation. LSHADE-cnEpSin [29] adds covariance matrix learning to LSHADE-EpSin. LSHADE-SPACMA [30] is hybrid of LSHADE and CMA-ES. ELSHADE-SPACMA is enhanced version of LSHADE-SPACMA. These variants introduce modifications like adaptive parameter control, linear population size reduction, and hybridization to improve DE's exploration/exploitation balance, convergence speed, and solution diversity. The primary goal of adaptive DE aims to aid in the exploration and exploitation of relationships that prevent early convergence difficulties and to optimize the final findings. Differential evolution with self-adapting populations [31] dynamically modifies the crossover and mutation parameters as well as the population count. Fuzzy adaptive differential evolution [32] is a form of DE method that adjusts the controller settings for crossover and mutation processes using fuzzy logic controllers. The mutation strategies "DE/rand/1" along with "DE/

current-to-best/2” are both subjected to self-adaptive differential evolution [33]. jDE is a comparable adaptive DE method to the conventional DE/rand/1/bin approach. Based on the modified parameters, JDE enhances the population count during the optimization process, resulting in vectors which are more probable to survive. The MDE employs a single array, that is updated whenever a superior approach is discovered. As a result, continually updating the same array improves convergence time, resulting in fewer evaluation processes than with traditional DE. Modified DE with P-Best Crossover uses F and Cr values generated by a Cauchy distribution with a position parameter, which are then modified on the power average of all F/Cr ratios generating successful offspring [34]. JADE is an option of modifying the parameters during each generation in order to achieve progressive self-adaptation depending on success rate [26]. Differential Covariance Matrix Adaptation Evolutionary Algorithm for Real Parameter Optimization was suggested by Saurav et al. The covariance matrix adaptation, like a quasi-Newton approach, aims to estimate the reverse Hessian matrix. Furthermore, the greedy selection approach of DE is used to enhance individuals in the following generation to boost the utility of the said approach [35]. Population initialization becomes a critical activity since it influences the quality of the eventual solution as well as the algorithm’s convergence time. The choice of this nonrepeatable procedure is critical, since the level of accuracy of the initial population collected influences the entire flow of the DE algorithm. Rahnamayan et al. [36] proposed an initialization approach based on opposition-based learning for producing the initial population of candidate solutions. Ozer et al. [37] suggested a chaotically initialized DE method that has a quicker convergence rate and is more resilient to premature convergence. Melo et al. [38] introduced smart sampling DE, which can discover the potential solution region in search arena by using the potency of machine learning. Zhu et al. [39] devised an adaptive population tuning system to dynamically regulate the population count of DE depending on the anticipated population dispersion as well as searching status. Cluster-based population initialization sampling strategy for DE was introduced by Poikolainen et al. [40] to anticipate the most feasible regions under decision space.

The current study proposes a self-adaptive DE algorithm based on the inherent features of the LSHADE-cnEpSin algorithm. Their major modifications are added to improve an algorithm’s overall performance. The CR is first adapted by employing oscillating iw , then the F is adapted employing SA-based modification, and lastly shrinking adaptation is used to lower the population count throughout the course of the rounds. The current work employs LSHADE as the core method, as well as SA-based changes, in addition to simple sinusoidal waves, have been performed to successfully pick a suitable F parameter. Proposed improvements attempt to add adaptive qualities to the algorithm to ensure no user-oriented alterations are necessary. The proposed approach is called LSHADE with SP adaptations (LSHADESPA). The work is the extended version of the paper accepted in ISCFI 2022 conference [41].

In this paper, the effectiveness of the LSHADESPA algorithm has been tested by utilizing renowned CEC 2014 [42], CEC 2017 [43], CEC 2021 [44], and CEC 2022 [45] test functions. A comparison is conducted between popular and recently introduced algorithms, such as i Laplacian BBO (LX-BBO) [46], population-based incremental learning (PBIL) [47], blended biogeography-based optimization (B-BBO) [46], Kepler optimization algorithm (KOA) [48], variable neighborhood bat (VNBA) [47], improved symbiotic organisms search (ISOS) [49], Self-adaptive DE (SaDE) [50], evolutionary algorithms with eigen crossover (EA4eig) [51], adaptive differential evolution with optional external archive (JADE) [26], success history-based adaptive DE (SHADE) [52], improved elephant herding optimization (IMEHO) [47], LSHADE [53], mean-variance mapping optimization (MVMO) [54], enhanced versions of CS (CV1.0) [55], improved version of CS (CV_{new}) [56], random walk gray wolf optimizer (RW-GWO) [57], DE [58], light spectrum optimizer (LSO) [59], chaotic cuckoo search (CCS) [47], PSO [58, 60], and nonlinear population size reduction success-history adaptive DE with rank-based selective pressure midpoint of the population (NL-SHADE-RSP-MID) [61]. The statistical significance of the suggested algorithm has been assessed employing two tests: the Wilcoxon rank-sum test as well as the Friedman test [62–64]. The main contribution of the paper is:

- A new variant of the DE algorithm utilizing three modifications (proportional SP ; SA-based scaling factor as well as oscillating iw-based crossover rate) known as the LSHADESPA algorithm is proposed.
- The LSHADESPA algorithm has been evaluated on the CEC 2014 [42] CEC 2017 [43], CEC 2021 [44], and CEC 2022 [45, 65] benchmarks test suite and compared with the other algorithms.
- Two statistical tests, including the Wilcoxon rank-sum as well as Friedman tests, have been performed to establish the statistical significance of the LSHADESPA algorithm.

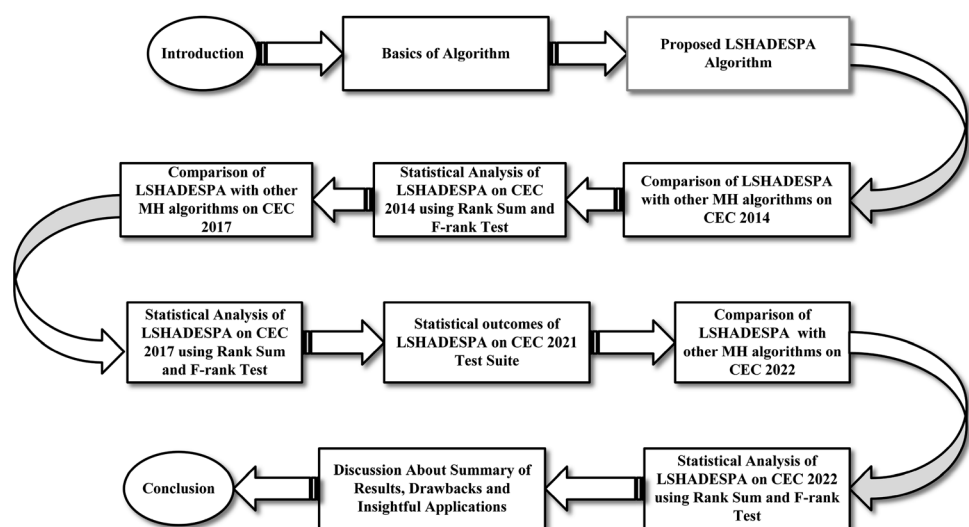
The entire paper is divided into six sections. The introduction section provides background information and key ideas, while Sect. 2 covers the fundamentals of DE algorithm. Section 3 begins with the proposal's general how's as well as why's while going into further depth about each change that has been made to the proposed LSHADESPA algorithm. This part also explains the justification for the proposal and provides theoretical information on how each modification increases the proposal's importance. Section 4 deals with results along with detailed discussions. Section 5 presents the summary of results, drawbacks, and insightful implication, whereas conclusions are being highlighted in the last section. The organization of the article is presented in Fig. 1.

2 Basics of DE

2.1 DE algorithm

Differential evolution is a population-based stochastic optimization algorithm used for optimizing real-valued nonlinear as well as nondifferentiable OF 's. It works by iteratively improving candidate solutions based on evolutionary principles of mutation, crossover, as well as selection. The key steps in differential evolution are: (1) Initialization—Generate an initial population of candidate solutions randomly. (2) Mutation- For each candidate solution, mutate it by adding the weighted difference of two random distinct candidates to a third one. (3) Crossover—Crossover the mutated candidate with the original candidate solution to generate a trial candidate. (4) Selection—Evaluate the trial candidate. If it is better than the original candidate, replace the original with the trial candidate. Repeat steps 2–4 until convergence or maximum iterations. The main parameters in differential evolution algorithm are: (a) Population size (NP)—Number of candidate solutions in a population. Typical values are 5–10 times the number of dimension (D). (b) Mutation factor (F)—Scales the difference vectors during mutation. Typical range is 0.5 to 1. (c) Crossover probability (CR)—Probability that trial vector inherits parameter values from the mutant. Typical range is 0.8–1. Mutation generates new vectors by adding the

Fig. 1 Organization of the article



weighted difference of two randomly selected candidates to a third one. Some common mutation strategies are: (a) DE/rand/1—Randomly pick three candidates (x_1, x_2, x_3), generate mutant as $x_3 + F * (x_1 - x_2)$ (b) DE/best/1—Pick best candidate as base x_3 , generate mutant as $x_3 + F * (x_1 - x_2)$ Mutation expands search space and enhances diversity in the population. Crossover mixes the mutated candidate with the original candidate to generate a trial candidate. Binomial crossover mixes parameters at random based on a crossover probability CR. Exponential crossover starts from the left and copies consecutive mutant parameters until a random number exceeds CR. Crossover increases diversity of candidates and avoids premature convergence. The trial candidate competes against the original candidate. If the trial candidate has better fitness, it replaces the original candidate in the population. This greedy selection pressure pushes the population toward optimal solutions. In summary, differential evolution is a simple yet powerful population-based direct search algorithm for global optimization. It is easy to implement, robust, fast, and requires minimal parameter tuning. With mutation, crossover, and selection, it can solve complex real-world optimization problems efficiently.

An initial population of n solution vectors is generated for a d -dimensional optimization problem with d parameters. Every generation t has a unique set of solutions, denoted by a_i , for which $i = 1, 2, \dots, n$.

$$a_i = a_{1,i}^t, a_{2,i}^t, \dots, a_{d,i}^t, \quad (1)$$

This vector can be considered the chromosomes or genomes. The mutation scheme is used to carry out the mutation. We first randomly select three different vectors (a_p , a_q , and a_r) for each vector a_i at any time or generation t . Then, the mutation process generates a so-called donor vector.

$$v_i^{t+1} = a_p^t + F(a_q^t - a_r^t) \quad (2)$$

where $F \in [0, 2]$ is a parameter, often referred to as the differential weight. This requires that the minimum number of the population size is $n \geq 4$.

The crossover is controlled by a crossover parameter $CR \in [0, 1]$, controlling the rate or probability for crossover. The process of crossover can be conducted using two distinct methods: binomial as well as exponential. The binomial system executes crossover operations on each of the d components or variables/parameters. By generating a uniformly distributed random number $r_i \in [0, 1]$, the j th component of v_i is formulated as

$$u_{j,i}^{t+1} = \begin{cases} v_{j,i} & \text{if } r_i \leq CR, \\ a_{j,i}^t & \text{Otherwise} \end{cases} \quad (3)$$

Algorithm 1 Differential evolution algorithm

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Initialize population  $x$  with randomly generated solutions;
Set the weight  $F \in [0, 2]$  and Crossover probability  $C_r \in [0, 1]$ 
if ( $i = 1 : t_{max}$ ) then
    for  $i = 1 : n$ 
        For each  $a_i$ , randomly select 3 vectors  $a_p$ ,  $a_q$ , and  $a_r$ 
        Produce a new vector  $v$ 
        Produce a random number  $r_i \in [0, 1]$ 
        for  $j = 1 : d$ 
            For each parameter  $v_{j,i}$ , update

$$u_{j,i}^{t+1} = \begin{cases} v_{j,i} & \text{if } r_i \leq C_r, \\ a_{j,i}^t & \text{Otherwise} \end{cases}$$

            Select and Update the solution
        end if
    end if
Post-process as well as output the best solution found

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3 Proposed approach

In the following subsections, an detailed analysis of the proposal's requirements and the methodology utilized for designing the proposed algorithm is presented.

3.1 Requirement of the proposal

In regard to recent advancements in optimization algorithms, the challenge of determining the superiority or inferiority of particular algorithms has become a regular and significant challenge. The No Free Lunch theorem (NFL) has been recognized in the discipline, demonstrating that the utilization of a single optimization algorithm cannot be generally successful for estimating the performance of all research challenges within various domains [66]. In the present work similar adoption of DE algorithm is followed. The importance of this proposal comes from the fact that while DE algorithms have excellent qualities. But the DE algorithm shows deficiencies in terms of their exploitation and exploration phases, respectively. Therefore, this process results in the occurrence of local optimal solutions, and this restricts getting to of the global optimum. In order to address the challenges mentioned above, a new version of the DE algorithm known as LSHADESPA algorithm has been proposed. The LSHADESPA algorithm incorporates three significant modifications to enhance its performance. Firstly, a proportional *SP* mechanism is employed to decrease the computational cost. Secondly, a SA-based scaling factor (F) is introduced to enhance the exploration properties of the algorithm. Lastly, an oscillating iw-based CR is utilized to achieve a balance between exploitation as well as exploration.

In summary, the LSHADESPA algorithm has undergone multiple modifications to enhance its operational efficiency. The subsequent subsection provides an overview of the way in which each modification has been incorporated.

3.2 The proposal

The proposal section provides a brief overview of the novel proposed algorithm known as LSHADESPA algorithm. The LSHADE algorithm has been enhanced with adaptive properties through the implementation of three key modifications. These include the integration of oscillating iw-based CR, the utilization of SA-based

mutation factor F , and the incorporation of an adaptive SP strategy [41]. The algorithm commences by initializing the variable N within a confined search space, as specified by Eq. (4).

$$x_{t,0}^i = x_{\min}^i + a(x_{\max}^i - x_{\min}^i) \quad 10pt i = 1, 2, \dots, D \quad (4)$$

For t th search agent, the parameter value for a D -dimensional problem is denoted by $a \in [0, 1]$, while $x_{\min}$ and $x_{\max}$ represent the lower as well as upper limits, respectively, of the problem under consideration.

Following the initialization process, the next phase involves the execution of the mutation operation. In the present scenario, the mutation strategy employed is derived from the JADE algorithm. This strategy follows to the current-to-best strategy, as outlined in Eq. (5)

$$x_{t+1}^g = x_t^g + F_t^g \cdot (x_{pbest}^g - x_t^g) + F_t^g \cdot (x_{r1}^g - x_{r2}^g) \quad (5)$$

The trial vector, denoted as x_{t+1}^g , is derived through mutation from the population size N . The solution x_{r1}^g is a randomly selected solution from the mutated trial vector. The solution x_{pbest}^g represents the personal best solution for the g th iteration. Finally, the solution x_{r2}^g is obtained from an external archive A , which contains solutions from previous inferior parents. In the initial phases, the archive is comprised of a set of x_t^i individuals. But following each generation g , a selection process is implemented whereby random individuals are removed from the overall population. This is done to create space for the inclusion of new individuals from the archive.

To tackle the decrease in population size, a strategy known as the SP strategy, as outlined in reference [67], is implemented. In the framework of a generalized algorithm, it is observed that an initial phase of increased exploration is needed, while in the later stages, a shift toward adaptive exploitation becomes imperative. The algorithm benefits from reducing the population size as the number of iterations increases, as the algorithm effectively decreases the total number of search agents needed for carrying out the operation. In simpler term, the process of exploration needs a greater number of search agents in order to investigate all feasible locations inside the search space. On the other hand, exploitation can be achieved with a restricted number of search agents due to the concentration of the optimal solution within specific portions. Therefore, it is necessary to modify the population in order to effectively explore the search space in the early phases by utilizing a larger population. Following that, the population should converge toward a final solution through exploitation, which can be achieved by utilizing a smaller population.

In the present work, a phenomena of decreased population that is proportionally influenced by a study [67] are utilized. This methodology employs the principle of population reduction through an enhancement in fitness. The adoption of this algorithm is justified by the need to address multi-modal challenges, which require solutions that possess the ability to effectively explore extensive solution areas. Therefore, in the scenario where the initial population size is significantly large, it is possible for new solutions to thoroughly explore the whole search space, as well as over the course of iterations, these solutions may gradually converge toward the global best solution, which is located in a particular direction. Therefore, it becomes feasible to decrease the population size. The decreased population size facilitates greater genetic drift as well as helps in the exploration of new solutions while keeping the optimal solutions. In the last phases, every individual within the population is given an equitable opportunity to attain the status of the global optimal solution. In a generalized case, the equation that represents the decrease in population is presented in Eq. (6).

$$N_{t+1} = P \begin{cases} (1 - \Delta bf_t^{\text{best}})P_t, & \text{if } \Delta bf_t^{\text{best}} \leq \Delta bf_{\max}^{\text{best}} \\ (1 - \Delta bf_{\max}^{\text{best}})P_t, & \text{if } \Delta bf_t^{\text{best}} > \Delta bf_{\max}^{\text{best}} \\ \min_P, & \text{if } P_{t+1} > \min_N \end{cases} \quad (6)$$

where the population size at generation t is denoted as P_{t+1} , and the value of $\Delta bf_t^{\text{best}}$ is determined by $(\frac{bf_{t-1}^{\text{best}} - bf_{t-2}^{\text{best}}}{|bf_{t-2}^{\text{best}}|})$,

which represents the change in the best fitness, and threshold value is denoted by $\Delta bf_{\max}^{\text{best}}$. The minimum population is defined in order to reduce the effect of a small N . The population curve shape is an exponential pattern during its early stages, maintaining a consistent trajectory during the middle phase, and once again changing to an exponential growth pattern toward the final stages. It is important to observe that the mentioned curve pattern is consistently observed throughout successive iterations, unless a minimum population size is reached or termination criteria are met. The utilization of an ensemble of sinusoidal adapted F is employed, obtaining inspiration from the LSHADE-cnEpSin algorithm. In this work, two sinusoidal decreasing adjustments are outlined in the given Eqs. (7) and (8).

$$F_i^{y+1} = \frac{1}{2} \times \left(\sin(2\pi \times \text{freq} \times y + \pi) \times \frac{y_{\max} - y}{y_{\max}} + 1 \right); \text{ if } r_1 > 0.5 \quad (7)$$

$$F_i^{y+1} = \frac{1}{2} \times \left(\sin(2\pi \times \text{freq} \times y) \times \frac{y_{\max} - y}{y_{\max}} + 1 \right); \text{ if } r_1 < 0.5 \quad (8)$$

In the present configuration, the freq is further adjusted through the utilization of SA iw's, as described in reference [68]. This adjustment is mathematically formulated as follows Eq. (9):

$$\text{freq} = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \times p^{(j-1)} \quad (9)$$

where the value of $\alpha_{\max}$, $\alpha_{\min}$ as well as $j \in [0, 1]$ and value of $p = 0.95$. The utilization of an iw adaptation is mainly inspired by the enhanced convergence characteristics shown by the SA approach. In addition, this approach helps in the production of larger step sizes during the first stage, which is a critical factor in facilitating more effective exploration. On the other hand, as the process progresses, it generates smaller steps, which promotes improved exploitation operations. In addition, exploration as well as exploitation, CR is additionally carried out through the utilization of the oscillating iw technique [68, 69]. The mathematical expression for oscillating iw can be represented as following Eq. (10):

$$CR = \frac{\beta_{\min} + \beta_{\max}}{2} + \frac{\beta_{\min} - \beta_{\max}}{2} + \cos \frac{2\pi g}{A} \quad (10)$$

$$A = \frac{2S_1}{3 + 2j} \quad (11)$$

where the value $\beta_{\max} = 0.9$, $\beta_{\min} = 0.5$, S_1 and $j \in [0, 1]$, respectively. The utilization of this iw is significantly advantageous as it facilitates the algorithm's smooth transition from the exploration to the exploitation phase while maintaining optimal speed and avoiding potential trapping in local minima. This allows the algorithm to show a gradual convergence throughout its earlier stages, followed by a faster convergence in its final stages. Note that after every generation, the archive or memory experiences update to allow for the storage of information related to previous generations. The algorithm for the proposed algorithm is given in Algorithm 2.

Algorithm 2 LSHADESPA algorithm

$G = 1$, $N_G = N_{\text{init}}$, Archive $A = \emptyset$
 Initialize population $P_G = (x_{1,G}, \dots, x_{N,G})$ randomly
 Set all values in M_{CR} , M_{F} to 0.5
while The termination criteria are not met **do**
 $S_{\text{CR}} = \emptyset$, $S_{\text{F}} = \emptyset$
 for $i = 1$ to N **do**
 $r_i = \text{Select from } [1, H] \text{ randomly}$
 if $M_{\text{CR}, r_i} = \perp$ **then**
 $CR_{i,G} = 0$
 else
 $CR_{i,G} = \text{randn}(M_{\text{CR}, r_i}, 0.1)$
 end if
 $F_{i,G} = \text{randc}(M_{\text{F}, r_i}, 0.1)$
 if $r_1 > 0.5$ **then**
 Generate F_i using Eq. (7)
 else
 Generate F_i using Eq. (8)
 end if
 Generate trial vector $u_{i,G}$ according to current-to-pbest/1/bin as in Eq. (5)
 end for
 for $i = 1$ to N **do**
 if $f(u_{i,G}) \leq f(x_{i,G})$ **then**
 $x_{i,G+1} = u_{i,G}$
 else
 $x_{i,G+1} = x_{i,G}$
 end if
 if $f(u_{i,G}) < f(x_{i,G})$ **then**
 $x_{i,G} \rightarrow A$
 $CR_{i,G} \rightarrow S_{\text{CR}}$, $F_{i,G} \rightarrow S_{\text{F}}$
 end if
 end for
 If necessary, delete randomly selected individuals from the archive such that the archive size is $|A|$
 Update memories M_{CR} and M_{F} (Algorithm 3)
 Calculate N_{G+1} according to Eq. (6)
 if $N_G < N_{G+1}$ **then**
 Sort individuals in P based on their fitness values and delete lowest $N_G - N_{G+1}$ members
 Resize archive size $|A|$ according to new $|P|$
 end if
 $G++$
end while

Algorithm 3 Memory update algorithm in SHADE

```

if  $S_{CR} = \emptyset$  and  $S_F = \emptyset$  then
    if  $M_{CR,k,G} = \perp$  or  $\max(S_{CR}) = 0$  then
         $M_{CR,k,G+1} = \perp$ 
    else
         $M_{CR,k,G+1} = \text{meanWL}(S_{CR})$ 
    end if
     $M_{F,k,G+1} = \text{meanWL}(S_F)$ 
     $k++$ 
    if  $k > H$  then
         $k = 1$ 
    end if
else
     $M_{CR,k,G+1} = M_{CR,k,G}$ 
     $M_{F,k,G+1} = M_{F,k,G}$ 
end if

```

In Algorithm 3, the index k ($1 \leq k \leq H$) determines which position in the memory should be updated. At the start of the search, k is set to 1. Each time a new element is added to the history, k is incremented. If k exceeds H , it is reset to 1.

In the update process, all individuals in generation G fail to produce a trial vector better than the parent, i.e., $S_{CR} = S_F = \emptyset$, the memory is not updated.

The weighted Lehmer mean $\text{meanWL}(S)$ is calculated using the formula below. The fitness improvement Δf_k influences the parameter adaptation (S refers to either S_{CR} or S_F).

$$\text{meanWL}(S) = \frac{\sum_{k=1}^{|S|} w_k \cdot S_k^2}{\sum_{k=1}^{|S|} w_k \cdot S_k} \quad (7)$$

$$w_k = \frac{\Delta f_k}{\sum_{l=1}^{|S_{CR}|} \Delta f_l} \quad (8)$$

$$\Delta f_k = |f(u_{k,G}) - f(x_{k,G})| \quad (9)$$

When updating M_{CR} , if $M_{CR,k,G} = \perp$ (where $\perp$ is a special “terminal value”) or $\max(S_{CR}) = 0$ (meaning all elements of S_{CR} are 0), then $M_{CR,k,G+1}$ is set to $\perp$. If M_{CR} is assigned the terminal value $\perp$, it remains fixed at $\perp$ until the end of the search. This locks CR_i to 0, enforcing a “change-one-parameter-at-a-time” policy, which slows down convergence but is effective for multimodal problems.

3.3 Computational complexity

To determine the computational complexity, the key steps are as:

1. Mutation & crossover (generating trial vectors):

- Each individual undergoes mutation and crossover using the *current-to-pbest/1/bin* strategy.
- Mutation requires the selection of the best individuals ($O(\log N)$, if sorted initially).
- The operations (vector additions, scaling, and binomial crossover) are $O(D)$.
- Total cost per individual: $O(D)$.
- Total cost per generation: $O(ND)$.

2. Selection & archive update:

- Each individual is compared with its trial vector.
- Function evaluations dominate: $O(D)$.
- Archive updates are $O(1)$.
- Total cost per generation: $O(ND)$.

3. Population resizing (occasionally):

- Sorting is at most $O(N \log N)$, but since it happens only when resizing occurs, it is not a dominant cost compared to $O(ND)$.
- Removing individuals is $O(N)$, which is negligible compared to $O(ND)$.

3.3.1 Final complexity

Since the dominant operations are mutation, crossover, and function evaluations, the per-generation complexity simplifies to:

$$O(ND)$$

Over G generations, the total complexity is:

$$O(GND)$$

This means that the run-time scales linearly with the number of generations G , population size N , and problem dimension D .

4 Results

This section presents an analysis of the performance of the proposed LSHADESPA algorithm across various benchmark suites. The purpose of this evaluation is to assess the performance of the LSHADESPA algorithm in comparison to other MH algorithms. The LSHADESPA algorithm is evaluated on the CEC 2014 [42] CEC 2017 [43], CEC 2021 [44] as well as CEC 2022 [45] benchmarks test suite. This section is divided into 4 subsection. In the first subsection, the performance of the LSHADESPA algorithm has been evaluated by utilizing CEC 2014 test suite and compared with the various MH algorithm such as IMEHO [47], LX-BBO [46], PBIL [47], B-BBO [46], VNBA [47], RW-GWO [57], ISOS [49], as well as CCS [47]. In 2nd subsection, the performance of the LSHADESPA algorithm is evaluated utilizing a set of 30 benchmark problems from the CEC 2017 and the outcomes of the LSHADESPA algorithm are compared with the other MH algorithms (CV1.0 [55], SaDE [50], MVMO [54], JADE [26], SHADE [52], LSHADE [53], SACS [70] as well as CV_{new} [56]). In the 3rd subsection, the LSHADESPA algorithm utilizes a set of 10 benchmark challenges from the CEC 2021 competition for further evaluation. In the last subsection, the performance of the LSHADESPA algorithm is tested with the CEC 2022 challenges and conducted a comparative analysis with other MH optimization algorithms (DE [58], LSO [59], KOA [48], PSO [58], EA4eig [51], and NL-SHADE-RSP-MID [61])

The algorithms were analyzed using a 12th generation, Windows 11 operating system, namely a 64-bit laptop with an i5-12,500 H, 16.00 GB of RAM, utilizing MATLAB version R2022a.

4.1 Statistical outcomes for CEC 2014 benchmarks

The proposed LSHADESPA algorithm performance is evaluated by using CEC 2014 test function. Various MH algorithms utilized for comparison are IMEHO [47], LX-BBO [46], PBIL [47], B-BBO [46], VNBA [47], RW-GWO [57], ISOS [49], as well as CCS [47]. The algorithms employed for comparison have been observed to be highly efficient in tackling complex optimization challenges.

Experimental testing: The simulations findings are presented in Table 1 in terms of error mean as well as std value. The table shows that functions F_1 , F_2 , F_3 , F_7 , as well as F_8 demonstrate that only the LSHADESPA algorithm is able to produce outcomes that are closely to the global optimal solution, while the remaining MH optimization techniques may become trapped in local minima. Also for functions F_4 , F_5 , F_6 , F_9 , F_{10} , F_{11} , F_{16} , F_{17} , F_{18} , F_{19} , F_{20} , F_{21} , F_{22} , F_{23} , F_{25} , F_{26} , and F_{20} , LSHADESPA algorithm shows superior performance while comparing with the other MH optimization techniques. Both the algorithms (B-BBO and LX-BBO algorithms) display similar outcomes in terms of error mean as well as std for the function F_{12} , and it is very highly difficult to comment which algorithm is good for this test function. For functions F_{13} , F_{14} , as well as F_{15} , IMEHO algorithm displays best outcomes. Also ISOS algorithms show superior outcomes as compared to other MH algorithms for function F_{24} . For the functions F_{28} , F_{29} , and F_{30} , the RW-GWO algorithm shows best performance as compared to the other MH algorithms. Figure 2 illustrates the radarchart of LSHADESPA, LX-BBO, B-BBO, IMEHO, CCS, RW-GWO, ISOS, PBIL, and VNBA for CEC 2014 numerical challenges.

Statistical testing: Two statistical tests, namely Wilcoxon's rank-sum [62] as well as the Friedman rank (f-rank) test [49], are utilized, to assess the statistical importance of the LSHADESPA algorithm. Both tests presented listed below are of a nonparametric type.

Wilcoxon's rank-sum test: This test is used to compare the performance of two algorithm and assign the p-rank. The rank of every test function is provided in the third row of Table 1 for each test function and shows in terms of win(w)/loss(l)/tie(t). The w condition will arise when the algorithm performs better than the proposed algorithm and assigns "+" sign. l condition will arise when the proposed algorithm performs better as compared to the other algorithm and assigns "-" sign. For the last condition, t will occur when both the algorithm performance is the same and assign "=" . So from Table 1 and Fig. 3 it is clearly seen that the proposed LSHADESPA algorithm shows superior performance as compared to the other MH algorithm.

Friedman rank test: Friedman rank (f-rank) test is also used to validate the performance of the LSHADESPA algorithm. In the f-rank test every algorithm assigned with unique rank, based on their performance, is presented in the 4th row of Table 1. The overall f-rank of every algorithm is presented in the last row of Table 1. From Table 1 and Fig. 4, it is clearly displayed LSHADESPA algorithm obtained the first rank.

4.2 Statistical outcomes for CEC 2017 benchmarks

In this subsection, the effectiveness of the LSHADESPA algorithm is evaluated utilizing a set of thirty benchmark challenges from the CEC 2017 competition. For CEC 2017 benchmark functions, the proposed LSHADESPA algorithm is contrasted with other MH algorithms, namely SaDE [50], JADE [26], SHADE [52], LSHADE [53], MVM0 [54], CV1.0 [55], SACS [70], and CV_{new} [56]. All of the algorithms chosen for comparison are extremely competitive as well as have proven their usefulness in solving [41]. The simulation outcomes for all algorithms are compared as well as presented in terms of the mean and standard deviation (std) values. From Table 2, it has been analyzed that for function pF_1 the performance of SHADE and LSHADE is optimum. For function,

Table 1 Comparison of the outcomes of LSHADESPA algorithm with other MH algorithms for CEC 2014

Functions		LSHADESPA	LX-BBO [46]	B-BBO [46]	IMEHO [47]	CCS [47]	RW- GWO [57]	ISOS [49]	PBIL [47]	VNBA [47]
F_1	Mean	0.000E+00	1.01E+07	6.50E+06	2.37E+06	1.46E+08	8.02E+06	9.82E+05	3.42E+08	2.43E+08
	Std	0.000E+00	1.01E+07	1.30E+06	4.32E+06	3.27E+07	3.31E+06	7.05E+05	1.09E+08	5.93E+07
	p- rank		–	–	–	–	–	–	–	–
	f-rank	1	6	4	3	7	5	2	9	8
F_2	Mean	0.000E+00	5.34E+04	2.35E+04	5.49E+03	2.60E+09	2.23E+05	5.27E+00	4.08E+10	1.92E+10
	Std	0.000E+00	2.14E+04	9.99E+03	4.87E+03	5.22E+08	5.51E+05	1.72E+01	3.39E+09	4.23E+09
	p- rank		–	–	–	–	–	–	–	–
	f-rank	1	5	4	3	7	6	2	9	8
F_3	Mean	0.000E+00	1.63E+04	6.03E+03	1.41E+02	2.70E+03	3.16E+02	4.79E+02	9.19E+04	2.93E+04
	Std	0.000E+00	1.70E+04	3.15E+03	1.58E+02	7.74E+04	4.34E+02	6.24E+02	1.75E+04	1.39E+04
	p- rank		–	–	–	–	–	–	–	–
	f-rank	1	7	6	2	5	3	4	9	8
F_4	Mean	4.7014E−05	9.99E+01	1.02E+02	3.41E+01	1.24E+02	3.22E+02	5.98E+01	3.43E+03	1.60E+03
	Std	3.3507E−04	2.84E+01	3.13E+01	4.77E+01	4.09E+01	1.80E+01	3.57E+01	7.56E+02	3.63E+02
	p- rank		–	–	–	–	–	–	–	–
	f-rank	1	4	5	6	7	3	2	9	8
F_5	Mean	2.0149E+01	3.06E+00	3.74E+00	2.10E+01	2.10E+01	2.05E+01	2.03E+01	2.10E+01	2.10E+01
	Std	3.2489E−02	7.86E−01	4.91E−01	5.99E−02	8.81E−02	7.46E−02	6.67E−02	5.56E−02	5.43E−02
	p- rank		–	–	–	–	–	–	–	–
	f-rank	1	9	8	4	7	6	5	3	2
F_6	Mean	5.7819E−06	1.70E+01	1.99E+01	1.20E+01	2.50E+01	9.84E+00	1.05E+01	3.80E+01	3.30E+01
	Std	1.9366E−05	3.12E+00	2.70E+00	2.72E+00	2.00E+00	3.49E+00	2.39E+00	1.16E+00	2.58E+00
	p- rank		–	–	–	–	–	–	–	–
	f-rank	1	5	6	4	7	2	3	9	8
F_7	Mean	0.000E+00	1.75E−01	7.81E−02	0.00E+00	2.30E+01	2.53E−01	1.56E−02	3.40E+02	1.11E+02
	Std	0.000E+00	8.56E−02	4.44E−02	1.19E−01	3.52E+00	1.43E−01	1.83E−02	2.74E+01	1.81E+01
	p- rank		–	–	–	–	–	–	–	–
	f-rank	1	5	4	2	7	6	3	9	8
F_8	Mean	0.000E+00	5.53E+01	4.71E−01	3.30E+01	2.90E+02	4.38E+01	1.47E+01	3.00E+02	1.74E+02
	Std	0.000E+00	3.78E+02	6.79E−01	9.19E+00	2.23E+01	8.48E+00	3.34E+00	1.03E+01	1.61E+01
	p- rank		–	–	–	–	–	–	–	–
	f-rank	1	6	2	4	8	5	3	9	7
F_9	Mean	1.2385E+01	7.66E+01	9.11E+01	3.20E+01	2.90E+02	6.33E+01	2.56E+02	3.70E+02	2.50E+02
	Std	2.6862E+00	1.61E+01	1.54E+01	1.15E+01	2.38E+01	1.30E+01	1.34E+01	1.69E+01	2.03E+01
	p- rank		–	–	–	–	–	–	–	–
	f-rank	1	4	5	2	8	3	7	9	6

Table 1 (continued)

Functions		LSHADESPA	LX-BBO [46]	B-BBO [46]	IMEHO [47]	CCS [47]	RW- GWO [57]	ISOS [49]	PBIL [47]	VNBA [47]
F_{10}	Mean	7.3333E−02	1.25E+04	6.68E+03	2.26E+03	8.55E+03	9.61E+02	1.78E+03	6.26E+03	3.50E+03
	Std	5.2993E−02	1.16E+02	4.58E+02	5.72E+02	4.91E+02	2.72E+02	4.09E+01	3.05E+02	3.47E+02
	p- rank		–	–	–	–	–	–	–	–
	f-rank	1	9	7	4	8	2	3	6	5
F_{11}	Mean	1.2785E+03	1.23E+04	6.71E+03	2.86E+03	8.83E+03	2.68E+03	1.48E+03	7.10E+03	6.80E+03
	Std	1.9627E+02	3.41E+02	5.17E+02	5.38E+02	5.50E+02	3.68E+02	4.54E+02	2.97E+02	3.79E+02
	p- rank		–	–	–	–	–	–	–	–
	f-rank	1	9	5	4	8	3	2	7	6
F_{12}	Mean	1.9835E−01	1.11E−02	1.11E−02	1.00E+01	1.00E+01	5.45E−01	3.55E−01	1.00E+01	1.00E+01
	Std	2.9275E−02	1.75E−18	1.75E−18	5.26E−01	1.09E+00	1.66E−01	5.73E−02	3.38E−01	3.51E−01
	p- rank		+	+	–	–	–	–	–	–
	f-rank	3	1	1	8	9	5	4	6	7
F_{13}	Mean	1.3820E−01	6.55E−01	6.78E−01	0.00E+00	0.00E+00	2.80E−01	3.77E−01	0.00E+00	0.00E+00
	Std	2.2746E−02	1.56v01	7.98E−02	6.25E−02	1.76E−01	6.30E−02	7.10E−02	2.56E−01	3.64E−01
	p- rank		–	–	–	+	+	–	–	–
	f-rank	3	8	9	1	2	6	7	4	5
F_{14}	Mean	1.8446E−01	6.20E−01	3.93E−01	0.00E+00	1.00E+01	4.23E−01	2.71E−01	1.00E+02	6.00E+01
	Std	2.6085E−02	2.96E−01	1.55E−01	9.85E−02	1.88E+00	2.15E−01	5.12E−02	1.16E+01	1.22E+01
	p- rank		–	–	+	–	–	–	–	–
	f-rank	2	6	4	1	7	5	3	9	8
F_{15}	Mean	2.3924E+00	1.55E+01	1.88E+01	0.00E+00	8.00E+01	8.81E+00	1.06E+01	6.84E+05	2.39E+03
	Std	2.5254E−01	5.49E+00	5.64E+00	1.35E+00	3.03E+01	1.51E+00	3.71E+00	2.85E+05	1.22E+03
	p- rank		–	–	+	–	–	–	–	–
	f-rank	2	5	6	1	7	3	4	9	8
F_{16}	Mean	8.5223E+00	1.08E+01	1.06E+01	2.00E+01	2.00E+01	1.03E+01	9.21E+01	2.00E+01	2.00E+01
	Std	3.9762E−01	5.84E−01	6.25E−01	7.64E−01	1.75E−01	6.11E−01	7.31E−01	2.12E−01	3.66E−01
	p- rank		–	–	–	–	–	–	–	–
	f-rank	1	4	3	8	5	2	9	6	7
F_{17}	Mean	1.5498E+02	1.49E+06	1.27E+06	7.69E+04	1.15E+07	5.71E+05	1.75E+05	9.74E+06	2.53E+06
	Std	1.0197E+02	9.34E+05	5.46E+05	8.38E+04	4.59E+06	4.10E+05	1.64E+05	2.79E+06	3.34E+06
	p- rank		–	–	–	–	–	–	–	–
	f-rank	1	5	4	9	6	3	8	7	2
F_{18}	Mean	6.9997E+00	2.89E+03	8.22E+02	3.30E+03	1.10E+08	6.52E+03	3.89E+03	6.16E+08	1.66E+08
	Std	2.5029E+00	4.27E+03	1.00E+03	3.52E+03	4.66E+07	4.63E+02	5.15E+03	1.68E+08	1.03E+08
	p- rank		–	–	–	–	–	–	–	–
	f-rank	1	3	2	4	7	6	5	9	8

Table 1 (continued)

Functions		LSHADESPA	LX-BBO [46]	B-BBO [46]	IMEHO [47]	CCS [47]	RW- GWO [57]	ISOS [49]	PBIL [47]	VNBA [47]
F_{19}	Mean	2.5506E+00	5.19E+03	7.81E+03	1.05E+01	4.00E+01	1.14E+01	7.79E+01	1.90E+02	1.20E+02
	Std	6.5981E-01	5.67E+03	4.67E+03	1.74E+00	5.91E+00	2.03E+00	1.78E+00	3.42E+01	3.82E+01
	p- rank		—	—	—	—	—	—	—	—
	f-rank	1	8	9	2	4	3	5	7	6
F_{20}	Mean	2.5130E+00	2.61E+04	1.62E+04	2.10E+02	1.03E+06	6.27E+02	4.98E+03	3.59E+04	1.69E+04
	Std	8.1871E-01	1.57E+04	4.11E+03	8.17E+01	9.05E+05	1.12E+03	3.40E+03	1.77E+04	6.57E+03
	p- rank		—	—	—	—	—	—	—	—
	f-rank	1	7	5	2	9	3	4	8	6
F_{21}	Mean	9.8109E+01	1.11E+06	1.22E+06	2.72E+04	5.66E+06	2.58E+05	8.90E+04	2.52E+06	2.30E+06
	Std	9.2975E+01	7.95E+05	7.96E+05	1.83E+04	2.73E+06	1.76E+05	1.07E+05	1.19E+06	1.35E+06
	p- rank		—	—	—	—	—	—	—	—
	f-rank	1	5	6	2	9	4	3	8	7
F_{22}	Mean	6.3215E+01	1.88E+03	1.68E+02	2.10E+02	1.34E+03	2.08E+02	2.75E+02	1.02E+03	8.40E+02
	Std	5.4478E+01	2.03E+02	2.47E+02	1.01E+02	1.88E+02	2.08E+02	1.45E+02	1.88E+02	1.28E+02
	p- rank		—	—	—	—	—	—	—	—
	f-rank	1	9	2	4	8	3	5	7	6
F_{23}	Mean	3.1507E+02	4.11E+02	3.43E+02	3.20E+02	3.50E+02	3.15E+02	3.15E+02	6.00E+02	3.90E+02
	Std	6.3290E-02	6.43E+01	2.84E+01	4.78E-01	7.66E+00	2.77E-01	1.60E+01	6.70E+01	2.47E+01
	p- rank		—	—	—	—	—	—	—	—
	f-rank	1	8	5	4	6	2	3	9	7
F_{24}	Mean	2.0977E+02	1.47E+04	3.41E+04	2.40E+02	2.20E+02	2.00E+02	2.00E+02	4.00E+02	2.30E+02
	Std	1.0903E+01	8.37E+03	2.35E+04	6.46E+00	2.51E+00	3.04E-03	1.50E-03	1.42E+01	2.53E+01
	p- rank		—	—	—	—	+	+	—	—
	f-rank	3	8	9	6	4	2	1	7	5
F_{25}	Mean	2.02553E+02	5.29E+02	6.53E+02	2.10E+02	2.20E+02	2.04E+02	2.00E+02	2.40E+02	2.10E+02
	Std	3.5990E-02	4.36E+01	6.01E+01	2.08E+00	4.44E+00	1.18E+00	8.07E-01	6.00E+00	1.07E+01
	p- rank		—	—	—	—	—	—	—	—
	f-rank	1	8	9	5	6	3	2	7	4
F_{26}	Mean	1.0013E+02	2.12E+00	3.64E+01	1.00E+02	1.00E+02	1.00E+02	1.00E+02	1.00E+02	1.00E+02
	Std	2.0795E-02	3.46E+00	5.62E+01	6.00E-02	2.06E-01	7.36E-02	9.55E-02	2.09E-01	4.30E-01
	p- rank		—	—	—	—	—	—	—	—
	f-rank	1	8	9	2	5	3	4	6	7
F_{27}	Mean	3.0196E+02	1.95E+02	3.04E+02	5.80E+02	5.30E+02	4.09E+02	5.43E+02	1.08E+03	1.29E+03
	Std	1.4002E+01	1.04E+02	1.60E+02	1.41E+02	7.92E+01	6.09E+00	1.36E+02	2.80E+02	3.30E+01
	p- rank		—	—	—	—	—	—	—	—
	f-rank	1	4	5	6	3	2	6	8	9
F_{28}	mean	8.3377E+02	1.94E+03	2.12E+03	9.70E+02	1.44E+03	4.34E+02	9.68E+02	1.39E+03	1.68E+03
	std	1.8931E+01	1.04E+02	4.44E+02	2.44E+02	9.46E+02	8.45E+00	4.12E+01	1.33E+02	2.33E+02
	p- rank		—	—	+	—	—	—	—	—
	f-rank	2	8	9	4	6	1	3	5	7

Table 1 (continued)

Functions		LSHADESPA	LX-BBO [46]	B-BBO [46]	IMEHO [47]	CCS [47]	RW- GWO [57]	ISOS [49]	PBIL [47]	VNBA [47]
F_{29}	Mean	7.2308E+02	1.98E+07	3.09E+07	1.21E+03	1.20E+06	2.14E+02	5.70E+05	5.70E+06	7.47E+06
	Std	1.2212E+01	3.95E+06	6.91E+06	2.16E+02	7.03E+05	2.37E+00	2.14E+06	3.33E+06	1.20E+06
	p- rank		–	–	–	–	+	–	–	–
	f-rank	2	8	9	3	5	1	4	6	7
F_{30}	Mean	1.1465E+03	6.95E+06	1.38E+07 4.08E+03	7.67E+04	6.69E+02	2.38E+05	1.49E+05	1.89E+05	
	Std	4.3062E+02	1.03E+07	1.08E+07	1.44E+03	3.37E+04	2.14E+02	1.10E+03	5.55E+04	1.03E+05
	p- rank		–	–	–	–	+	–	–	–
	f-rank	2	8	9	3	4	1	7	5	6
w//t		NA	0/30/0	0/30/0	0/30/0	0/30/0	4/26/0	1/29/0	0/30/0	0/30/0
Overall f-rank value		41	190	175	113	191	102	123	221	196
f-rank		1	6	5	3	7	2	4	9	8

$pF_2, pF_{11}, pF_{15}, pF_{18}$, and pF_{22} , performance of LSHADESPA is best among all tested MH algorithms. For pF_3 LSHADESPA performs equally like SHADE and LSHADE. For the function $pF_4, pF_{17}, pF_{23}, pF_{24}, pF_{25}, pF_{27}$, and pF_{28} functions, CV_{new} algorithm performed best performance as compared to the other MH algorithms. In the case of functions $pF_5, pF_7, pF_8, pF_{10}, pF_{20}, pF_{21}$, and pF_{29} , the outcomes obtained from the LSHADE algorithm display superior performance when compared to the other MH techniques. From the outcomes obtained, it is clear that JADE algorithms display superior performance for the function pF_6 . For functions F_9 , it can be seen that LSHADE and LSHADESPA algorithms demonstrate similar outcomes in terms of mean as well as std. On the other hand, the MVO algorithm comes as the optimal choice for the functions of pF_{12}, pF_{13} and pF_{25} . In the case of functions pF_{16}, pF_{19} and pF_{30} , it can be determined that the SACS algorithm provides the most optimal outcomes in terms of mean as well as std values. Figure 5 presents the radar of LSHADE, SACS, SHADE, SaDE, CV1.0, MVMO, JADE, CV_{new} , and LSHADESPA for CEC 2017 numerical challenges. Note that the proposed LSHADESPA algorithm is found to be superior than the other MH algorithm under comparison for CEC 2017 benchmark test suite

Statistical testing To validate the performance of the proposed LSHADESPA, statistical test, namely Wilcoxon's rank-sum [62] as well as Friedman rank (f-rank) test [49], is performed. This test is nonparametric in nature and utilized to test the statistical significance of the algorithm. First, the rank-sum test determines if the proposed LSHADESPA algorithm is significantly superior or not. LSHADESPA's performance is provided below in terms of win(w)/loss(l)/tie(t). The w condition happens if the algorithm under test outperforms the proposed approach and is represented by the "+" sign. In the l condition, the test algorithm performs worse than the LSHADESPA algorithm, as indicated by the "–" sign. The final t circumstance happens when there is no statistical difference between the algorithms that are under test and is represented by the "=" symbol. The $w//l/t$ row in Table 2 and Fig. 6 shows that the proposed LSHADESPA performs much better than other algorithms in the majority of circumstances. From the table it is clearly indicated that LSHADESPA wins over SHADE in 25 functions, CV1.0 in 29 functions, JADE in 26 functions, MVMO in 25 functions, CV_{new} in 25 functions, LSHADE in 14 functions, and over SACS in 25 functions out of total 30 Functions. LSHADESPA lashes out SaDE in all 30 functions. After the rank-sum test, the efficiency of the algorithm is tested by utilizing the f-rank test. In the fourth row of each algorithm of Table 2, a unique rank is given based on their outcomes. In the last

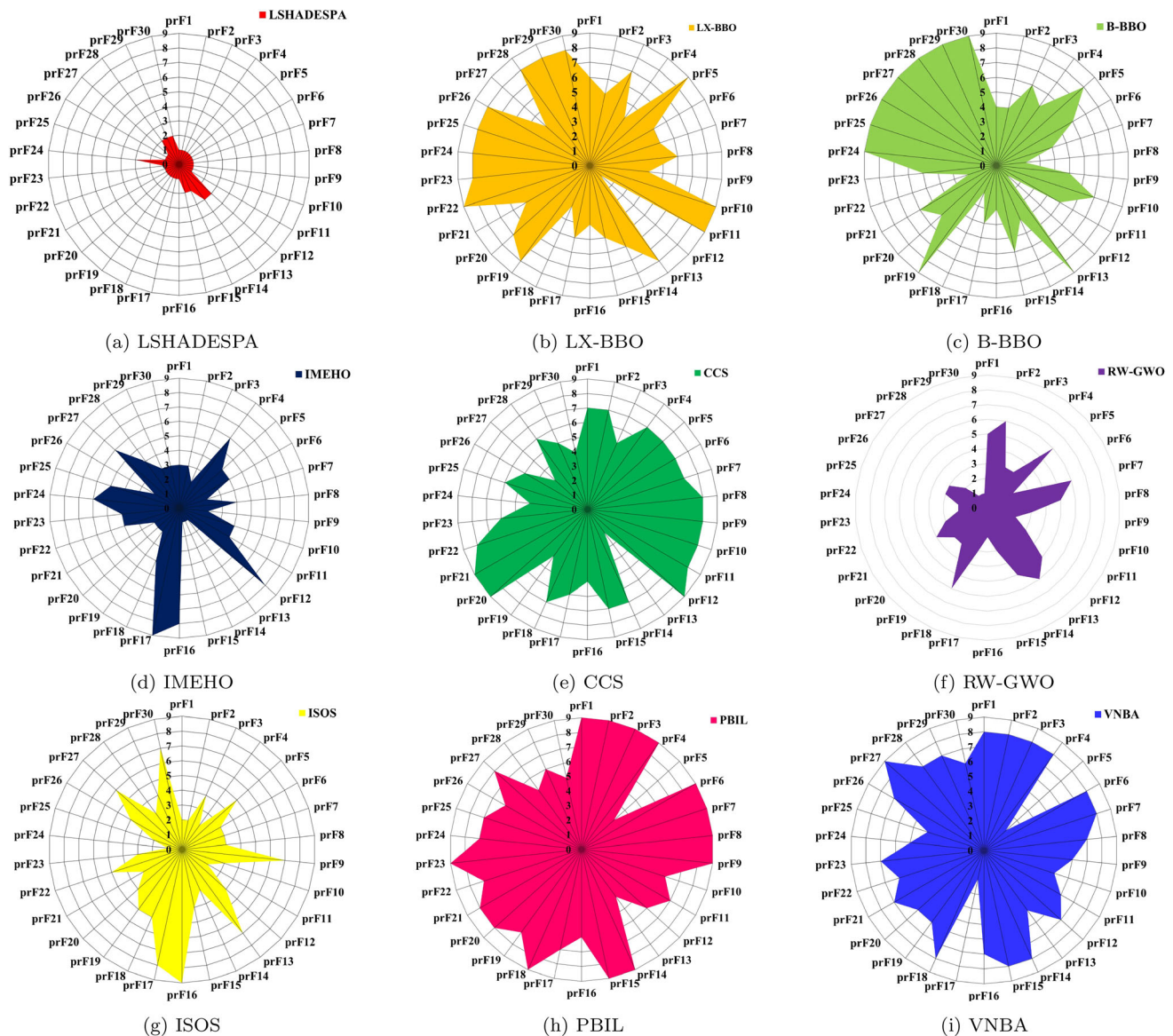


Fig. 2 Radarchart of LSHADESPA, LX-BBO, B-BBO, IMEHO, CCS, RW-GWO, ISOS, PBIL, and VNBA for CEC 2014 numerical challenges

row of the table f-rank value is given. From Table 2 and Fig. 7, it is clearly observed that the LSHADESPA algorithm got the 1st rank.

4.3 Statistical outcomes for CEC 2021 benchmarks

In this section, an evaluation is conducted to assess the performance of the LSHADESPA algorithm. The evaluation utilizes a set of 10 benchmark problems from the CEC 2021 competition. The minimization problems presented involve a series of operations, namely basic, rotation, translation, as well as shifting. These operations can be performed independently or in combination with one another? Additional detail on the CEC 2021 benchmark functions can be found in the reference [71]. The parametric settings utilized for the LSHADESPA algorithm for the CEC 2021 benchmarks are similar to those utilized in the CEC 2017 benchmarks. The experiments have been done a total of 30 times, with two different values for the variable $D = 10$ as well as 20.

Fig. 3 Comparison of the performance of the LSHA-DESPA algorithm with other algorithms for CEC 2014 (w//t scenario)

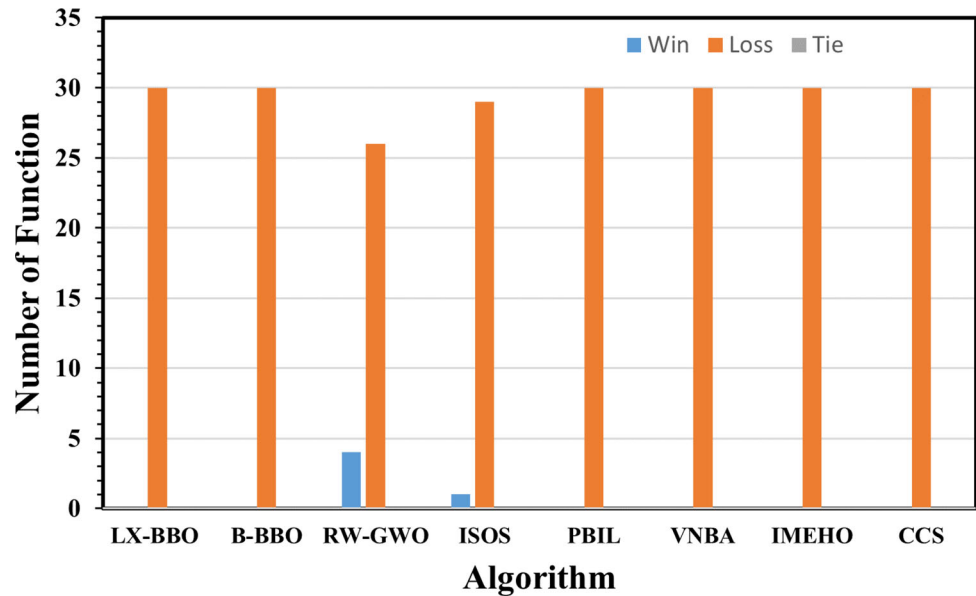
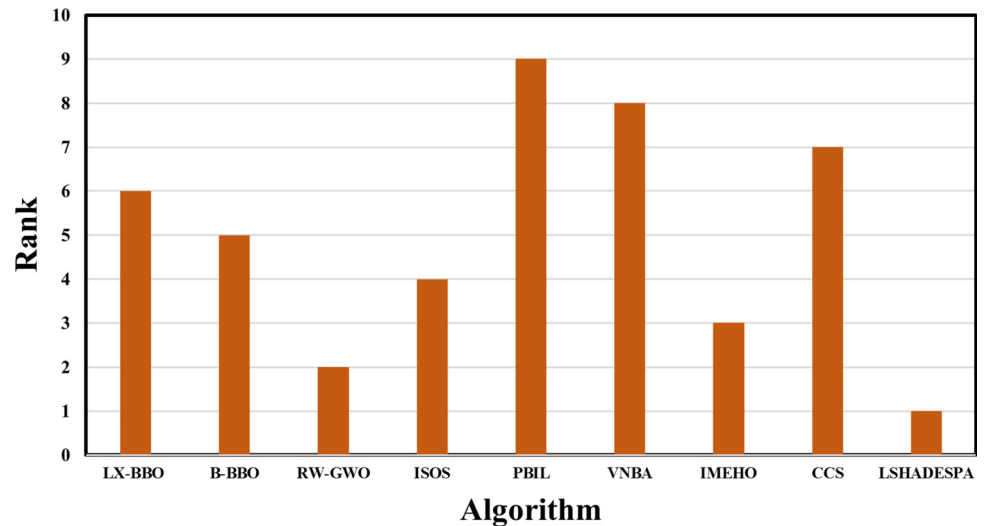


Fig. 4 Rankings of algorithms for CEC 2014



The maximum number of function evaluations for a problem with D equal to 10 is 200,000. However, for a problem with D equal to 20, the maximum number of function evaluations increases to 1,000,000. It is worth noting that each function in both cases has a search range of $[-100, 100]^D$. The outcomes obtained from the simulation for CEC 2021 are presented in Tables 3 and 4. The tables demonstrate that the proposed LSHA-DESPA algorithm displays an outstanding level of competitiveness. It consistently provides outcomes that are in closely matching to the optimal solutions for the CEC 2021 challenges.

4.4 Statistical outcomes for CEC 2022 benchmarks

As a further extension, the performance of the LSHADESPA algorithm is evaluated utilizing a set of 12 benchmark problems from the CEC 2022 competition in this subsection [45]. To determine the effectiveness of the LSHADESPA algorithm, a comparative analysis is conducted against other MH optimization algorithms. The algorithms employed for comparison include DE [58], LSO [59], KOA [48], PSO [58], EA4eig [51], and NL-SHADE-RSP-MID [61]. The outcomes of the CEC 2022 problem are presented in Table 5 in terms of mean and

Table 2 Comparison of the outcomes of LSHADESPA algorithm with other MH algorithms for CEC 2017

	LSHADE	SACS	SHADE	SaDE	CV1.0	MVMO	JADE	CV_{new}	LSHADESPA
pF_1	Mean	0.000E+00	0.000E+00	1.221E+03	1.000E+10	1.302E-05	5.201E-14	1.000E+10	6.901E-07
	Std	(0.000E+00)	(0.000E+00)	(1.940E+03)	(0.000E+00)	(5.571E-06)	(2.481E-14)	(0.000E+00)	(1.741E-06)
	p- rank	-	+	-	-	-	+	-	-
	f-rank	6	1	9	6	5	3	6	4
pF_3	Mean	0.000E+00	0.000E+00	2.701E+02	1.912E+04	5.262E-07	1.741E+04	8.681E+03	0.000E+00
	Std	(4.052E+03)	(0.000E+00)	(8.242E+02)	(6.241E+03)	(1.062E-07)	(3.671E+04)	(1.492E+03)	(0.000E+00)
	p- rank	=	=	-	-	-	-	-	-
	f-rank	6	1	5	8	4	9	7	1
pF_4	Mean	8.152E+01	4.921E+01	8.892E+01	3.552E+01	5.652E+01	1.121E+02	2.631E+01	3.592E+01
	Std	(4.792E+01)	(2.681E+01)	(4.692E+01)	(3.622E+01)	(8.792E+00)	(6.241E+03)	(5.891E+00)	(2.493E+01)
	p- rank	-	-	-	+	-	+	-	-
	f-rank	8	5	7	2	6	9	1	3
pF_5	Mean	1.182E+01	3.241E+01	9.222E+01	3.372E+02	8.042E+01	5.391E+01	2.361E+02	5.121E+01
	Std	(2.011E+00)	(5.011E+00)	(1.842E+01)	(8.051E+01)	(1.611E+01)	(8.762E+00)	(3.752E+01)	(1.411E+01)
	p- rank	+	+	-	-	-	-	-	-
	f-rank	1	2	6	9	5	4	8	3
pF_6	Mean	5.611E-05	8.351E-04	7.422E-03	4.861E+01	5.372E-03	1.411E-13	4.041E+01	1.501E-08
	Std	(3.682E-04)	(3.381E-01)	(2.311E-02)	(4.791E+01)	(3.272E-03)	(9.061E-14)	(8.101E+00)	(5.5210E-08)
	p- rank	-	-	-	-	+	-	-	-
	f-rank	3	4	6	9	5	1	8	2
pF_7	Mean	6.282E1+01	8.061E+01	1.412E+02	2.711E+02	1.201E+02	1.000E+02	2.191E+02	1.200E+02
	Std	(1.691E+00)	(4.032E+01)	(1.932E+01)	(7.232E+01)	(1.242E+01)	(6.451E+00)	(3.432E+01)	(1.771E+01)
	p- rank	+	+	-	-	-	+	-	-
	f-rank	1	2	6	8	5	3	6	4
pF_8	Mean	1.081E+01	5.492E+01	9.382E+01	7.542E+01	3.201E+01	3.261E+02	2.491E+02	5.381E+01
	Std	(2.242E+00)	(4.621E+01)	(1.761E+01)	(1.581E+01)	(3.781E+00)	(7.261E+01)	(4.472E+01)	(1.341E+01)
	p- rank	+	-	-	-	+	-	-	-
	f-rank	1	9	4	5	2	8	7	3

Table 2 (continued)

	LSHADE	SACS	SHADE	SaDE	CV1.0	MVMO	JADE	CV _{new}	LSHADESPA
pF_9	Mean	0.000E+00	4.741E+02	1.071E+00	1.000E+04	7.53E+00	1.131E+00	1.033E+04	0.000E+00
	Std	(0.000E+00)	(1.708E+03)	(9.347E-01)	(2.893E+03)	(5.745E+00)	(1.254E+00)	(3.091E+03)	(0.000E+00)
	p- rank	=	-	-	-	-	-	-	-
	f-rank	1	7	3	8	5	4	9	1
pF_{10}	Mean	3.143E+03	5.292E+03	3.311E+03	7.062E+03	3.463E+03	3.721E+03	6.061E+03	6.811E+03
	Std	(2.501E+02)	(6.093E+02)	(2.893E+02)	(5.301E+02)	(4.282E+02)	(2.513E+02)	(3.521E+02)	(3.461E+02)
	p- rank	+	+	+	-	+	+	+	-
	f-rank	1	5	2	9	3	4	6	7
pF_{11}	Mean	4.832E+01	6.551E+01	1.191E+02	1.633E+02	4.712E+01	1.311E+02	1.131E+02	3.241E+01
	Std	(7.881E+00)	(1.031E+01)	(2.901E+01)	(3.332E+01)	(8.693E+00)	(3.361E+01)	(1.862E+01)	(3.402E+00)
	p- rank	-	-	-	-	-	-	-	-
	f-rank	3	4	6	9	2	8	7	1
pF_{12}	Mean	2.132E+03	1.000E+10	5.102E+03	1.001E+10	1.261E+03	5.102E+03	1.000E+10	1.484E+03
	Std	(4.491E+02)	(0.000E+00)	(2.853E+03)	(0.000E+00)	(2.773E+02)	(3.282E+03)	(0.000E+00)	(4.888E+02)
	p- rank	-	-	-	-	+	-	-	-
	f-rank	3	7	4	9	1	5	7	2
pF_{13}	mean	6.232E+01	9.181E+09	2.601E+02	1.000E+10	4.341E+01	3.053E+02	9.792E+09	1.322E+02
	Std	(2.842E+01)	(2.732E+09)	(1.433E+02)	(0.000E+00)	(1.732E+01)	(2.641E+02)	(1.393E+09)	(3.121E+01)
	p- rank	+	-	-	-	+	-	-	-
	f-rank	2	7	4	9	1	5	8	3
pF_{14}	Mean	2.883E+01	2.644E+03	2.111E+02	2.011E+02	4.832E+01	1.011E+04	3.961E+01	4.185E+01
	Std	(2.902E+00)	(1.533E+02)	(7.266E+01)	(2.102E+01)	(1.184E+01)	(3.103E+04)	(1.581E+01)	(3.796E+00)
	p- rank	+	-	-	-	-	-	+	-
	f-rank	1	8	6	5	4	9	2	3
pF_{15}	Mean	4.142E+01	2.381E+01	3.201E+02	1.343E+09	4.412E+01	3.474E+02	2.821E+02	4.083E+01
	Std	(9.892E+00)	(5.933E+03)	(1.392E+02)	(3.443E+09)	(1.091E+01)	(4.392E+02)	(3.483E+02)	(6.181E+00)
	p- rank	-	+	-	-	-	-	-	-
	f-rank	4	1	6	9	3	7	5	2

Table 2 (continued)

		LSHADE	SACS	SHADE	SaDE	CV1.0	MVMO	JADE	CV _{new}	LSHADESPA
pF_{16}	Mean	3.732E+02	2.723E+02	7.301E+02	8.142E+02	1.502E+03	8.374E+02	8.533E+02	1.431E+03	7.192E+02
	Std	(1.142E+02)	(1.713E+02)	(1.821E+02)	(2.311E+02)	(2.753E+02)	(1.901E+02)	(1.712E+02)	(2.082E+02)	(1.271E+02)
	p- rank	+	+	-	-	-	-	-	-	-
	f-rank	2	1	4	5	9	6	7	8	3
	Mean	2.513E+02	5.012E+02	5.111E+02	5.075E+02	1.214E+03	5.163E+02	6.000E+02	1.101E+02	4.383E+02
pF_{17}	Mean	(7.421E+01)	(1.152E+02)	(1.081E+02)	(1.522E+02)	(1.824E+02)	(1.301E+02)	(1.185E+02)	(1.892E+02)	(9.761E+01)
	p- rank	+	-	-	-	-	-	-	+	-
	f-rank	2	4	6	5	9	7	8	1	3
	Mean	3.891E+01	1.032E+05	1.853E+02	3.232E+04	5.141E+02	4.123E+01	1.861E+02	1.482E+02	3.221E+01
	Std	(1.082E+01)	(4.161E+04)	(1.000E+02)	(1.676E+04)	(1.163E+02)	(1.912E+01)	(1.201E+02)	(4.403E+01)	(3.212E+00)
pF_{18}	p- rank	-	-	-	-	-	-	-	-	-
	f-rank	2	9	5	7	8	3	6	4	1
	Mean	2.423E+01	1.152E+02	1.561E+02	1.101E+04	1.721E+02	1.703E+01	3.232E+02	5.522E+01	2.771E+01
	Std	(8.783E+00)	(2.442E+00)	(5.633E+01)	(1.671E+04)	(4.142E+02)	(5.101E+00)	(1.203E+03)	(1.061E+01)	(1.131E+03)
	p- rank	-	+	-	-	-	+	-	-	-
pF_{19}	f-rank	4	1	6	9	7	2	8	5	3
	Mean	1.731E+02	3.231E+02	3.321E+02	3.462E+02	1.023E+03	3.241E+02	4.351E+02	2.782E+02	2.731E+02
	Std	(7.841E+01)	(8.632E+01)	(1.193E+02)	(1.511E+02)	(2.112E+02)	(1.442E+02)	(1.301E+02)	(1.623E+02)	(7.421E+01)
	p- rank	+	-	-	-	-	-	-	-	-
	f-rank	1	4	6	7	9	5	8	3	2
pF_{21}	Mean	2.092E+02	2.513E+02	2.301E+02	2.862E+02	5.383E+02	2.743E+02	2.501E+02	1.151E+02	3.201E+02
	Std	(1.901E+00)	(1.113E+02)	(5.082E+00)	(1.331E+01)	(6.222E+01)	(1.573E+01)	(9.601E+00)	(8.744E+01)	(1.444E+01)
	p- rank	+	-	-	-	-	-	-	-	-
	f-rank	1	4	6	7	9	5	8	3	2
	Mean	2.462E+03	1.000E+02	3.131E+03	2.912E+03	7.301E+03	3.232E+03	3.321E+03	5.711E+03	1.000E+02
pF_{22}	Std	(1.581E+03)	(6.072E+03)	(1.521E+03)	(3.212E+03)	(1.963E+03)	(1.688E+03)	(1.791E+03)	(3.632E+02)	(1.671E+03)
	p- rank	-	-	-	-	-	-	-	-	-
	f-rank	4	2	5	3	9	7	6	8	1

Table 2 (continued)

	LSHADE	SACS	SHADE	SaDE	CV1.0	MVMO	JADE	CV _{new}	LSHADESPA
pF_{23}	Mean	4.271E+02	5.111E+02	4.561E+02	5.211E+02	7.712E+02	4.763E+02	1.822E+02	4.712E+02
	Std	(5.033E+00)	(4.731E+01)	(8.722E+00)	(2.021E+01)	(8.033E+01)	(1.141E+01)	(5.083E+01)	(1.582E+01)
	p-rank	+	-	-	-	-	-	+	-
	f-rank	5	7	3	8	9	4	1	2
pF_{24}	Mean	5.042E+02	5.303E+02	5.301E+02	5.861E+02	8.292E+02	5.273E+02	3.221E+02	5.301E+02
	Std	(2.311E+00)	(2.623E+01)	(7.422E+00)	(1.873E+01)	(1.202E+01)	(7.591E+00)	(8.942E+01)	(1.472E+01)
	p-rank	-	-	-	-	-	-	+	-
	f-rank	7	3	5	8	9	4	1	2
pF_{25}	Mean	4.822E+02	4.801E+02	5.051E+02	5.701E+02	5.412E+02	5.183E+02	4.692E+02	4.792E+02
	Std	(1.602E+01)	(8.623E+00)	(3.611E+01)	(3.041E+01)	(1.482E+01)	(3.431E+01)	(2.231E+01)	(3.071E+00)
	p-rank	-	-	-	-	-	-	+	-
	f-rank	4	3	5	9	8	7	1	2
pF_{26}	Mean	1.111E+03	3.001E+02	1.381E+03	2.512E+03	2.423E+03	1.542E+03	1.132E+03	1.251E+03
	Std	(4.461E+01)	(2.382E-04)	(9.751E+01)	(3.362E+02)	(1.852E+03)	(1.181E+02)	(1.531E+03)	(6.882E+00)
	p-rank	+	+	-	-	-	-	-	-
	f-rank	2	1	5	9	8	6	4	3
pF_{27}	Mean	5.301E+02	5.242E+02	5.462E+02	7.132E+02	7.351E+02	5.491E+02	4.501E+02	5.142E+02
	Std	(1.862E+01)	(2.751E-01)	(2.712E+01)	(6.622E+01)	(8.221E+01)	(2.312E+01)	(7.123E+01)	(6.883E+00)
	p-rank	-	-	-	-	-	-	+	-
	f-rank	7	3	5	8	9	7	1	2
pF_{28}	Mean	4.701E+02	4.791E+02	4.731E+02	4.982E+02	4.911E+02	4.842E+02	4.552E+02	4.592E+02
	Std	(2.213E+01)	(7.532E-04)	(2.381E+01)	(1.501E+01)	(1.901E+01)	(2.051E+01)	(2.302E-01)	(1.133E+01)
	p-rank	-	-	-	-	-	-	+	-
	f-rank	4	6	5	9	8	7	1	2
pF_{29}	Mean	3.481E+02	1.541E+03	4.841E+02	5.134E+02	1.643E+03	4.714E+02	1.422E+03	5.5321E+02
	Std	(1.031E+01)	(1.491E+02)	(1.021E+02)	(1.361E+02)	(2.261E+02)	(8.051E+01)	(1.681E+02)	(3.381E+01)
	p-rank	+	-	-	-	-	-	-	-
	f-rank	1	8	4	6	9	2	7	5

Table 2 (continued)

		LSHADE	SACS	SHADE	SaDE	CV1.0	MVMO	JADE	CV_{new}	LSHADESPA
pF_{30}	Mean	6.501E+05	1.111E+05	6.811E+05	8.081E+05	4.631E+06	5.781E+05	6.671E+05	6.011E+05	6.131E+05
	Std	(7.251E+04)	(4.101E+04)	(8.481E+04)	(8.301E+04)	(8.561E+06)	(1.011E+04)	(9.221E+04)	(2.961E+04)	(3.961E+04)
	p- rank	–	+	–	–	–	–	–	–	–
	f-rank	6	1	8	9	2	3	7	4	5
$w//t$		13/14/2	5/24/0	4/24/1	0/29/0	1/28/0	5/24/0	4/25/0	8/21/0	NA
	Overall f-rank value	83	144	128	200	227	124	174	139	77
	f-rank	2	5	4	8	9	3	7	6	1

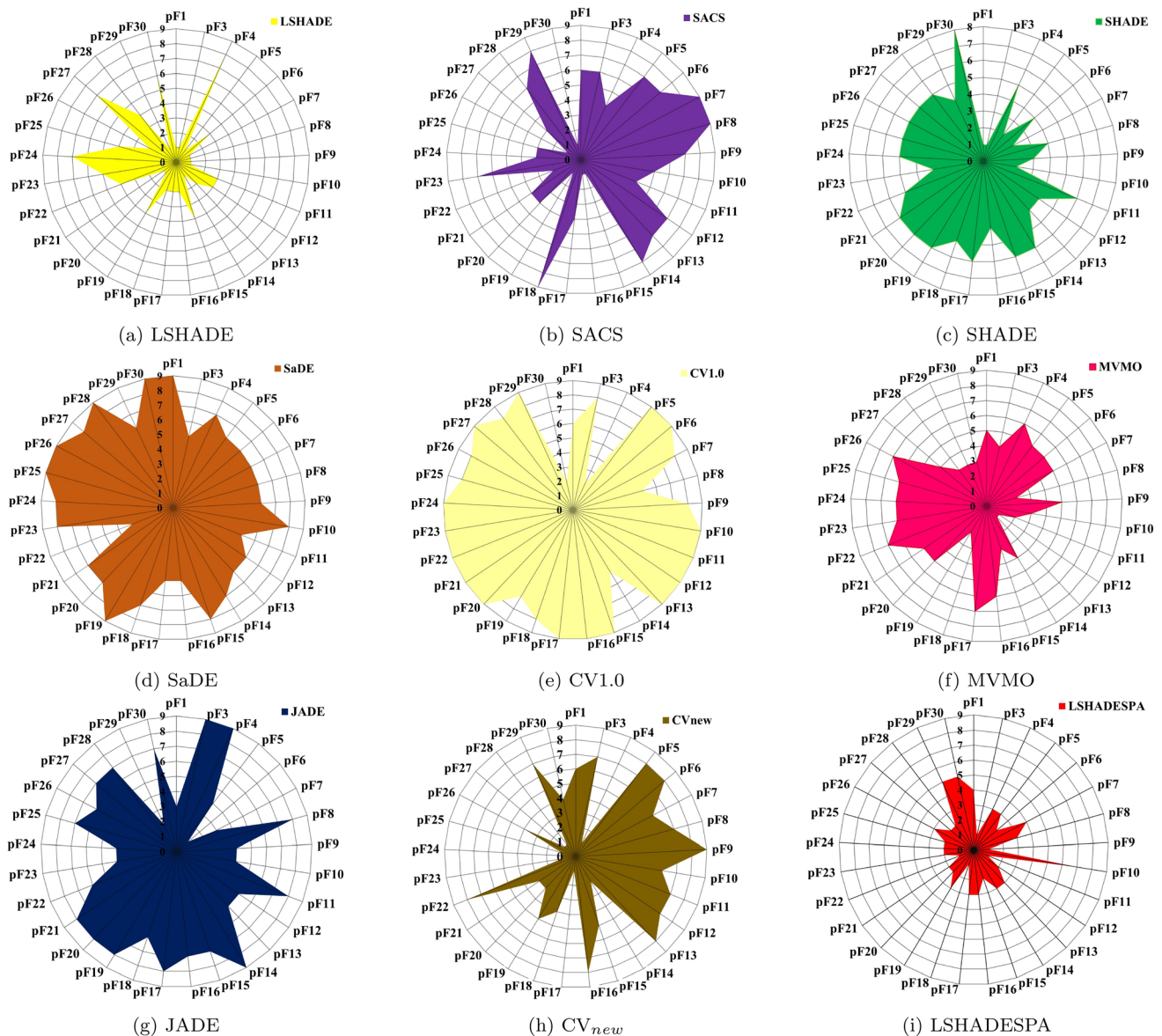


Fig. 5 Radarchart of LSHADE, SACS, SHADE, SaDE, CV1.0, MVMO, JADE, CV_{new}, and LSHADESPA for CEC 2017 numerical challenges

std for dimension size 10. For the problem, pr_1 , as well as LSHADESPA, LSO, NL-SHADE-RSP-MID algorithms, shows the same outcomes in terms of mean and std. For problem, pr_2 , NL-SHADE-RSP-MID algorithms show the superior performance. Also for the problem, pr_3 , LSHADESPA, DE, NL-SHADE-RSP-MID, as well as LSO algorithms demonstrate the same performance in terms of mean and std, and it is difficult to find which algorithm is best for this problem. EA4eig algorithm shows superior performance as compared to the other MH algorithm for problems pr_4 , pr_6 , as well as pr_9 . Both algorithms (LSHADESPA and NL-SHADE-RSP-MID) show the best outcomes for the problem pr_5 . For the problems pr_7 , pr_{10} , as well as pr_{11} , the LSHADESPA algorithm shows superior performance in terms of mean and std. NL-SHADE-RSP-MID algorithm shows the optimal result for the problem pr_8 . For the problem pr_{12} , LSO as well as LSHADESPA algorithms show exactly similar results in terms of mean value. Figure 8 shows the radarchart of LSHADESPA, DE, PSO, LSO, EA4eig, MVMO, NL-SHADE-RSP-MID, and KOA for CEC 2022 numerical challenges.

Fig. 6 Comparison of the performance of the LSHA-DESPA algorithm with other algorithms for CEC 2017 (*w/l/t* scenario)

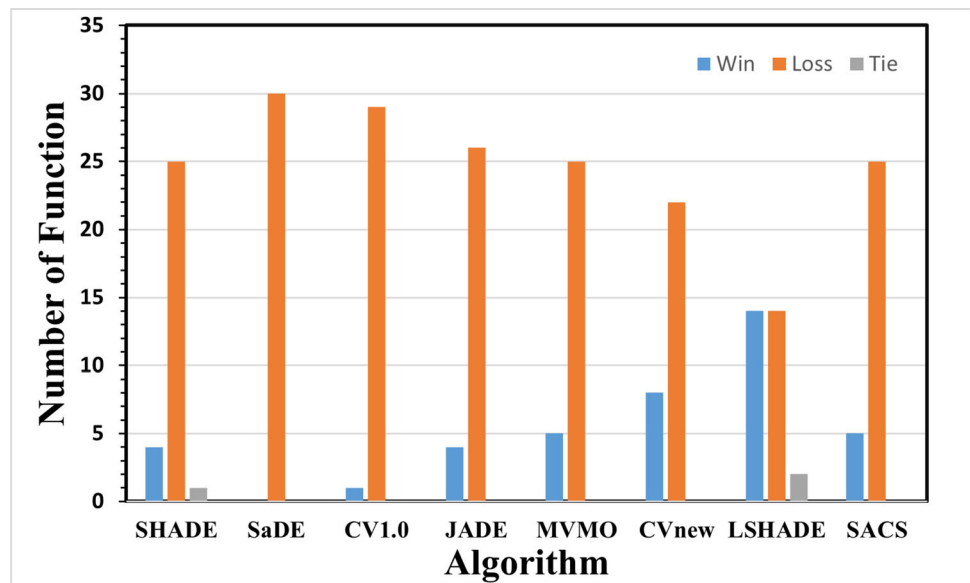
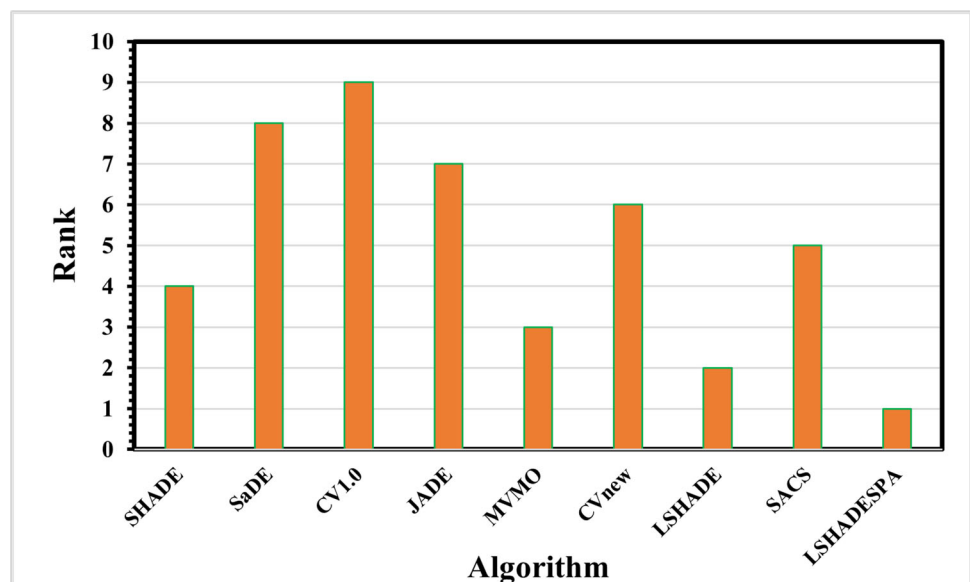


Fig. 7 Rankings of algorithms for CEC 2017



Statistical testing After the simulation outcomes, to prove the statistical importance of the proposed LSHA-DESPA algorithm, the rank-sum test is also performed. To validate the performance of the proposed LSHA-DESPA, statistical tests, namely Wilcoxon's rank-sum as well as Friedman rank (f-rank) test, are performed. This test is nonparametric in nature and utilized to test the statistical significance of the algorithm. First, the rank-sum test determines if the proposed LSHADESPA algorithm is significantly superior or not. LSHADESPA's performance is provided below in terms of win(*w*)/loss(*l*)/tie(*t*). The *w* condition happens if the algorithm under test outperforms the proposed approach and are represented by the “+” sign. In the *l* condition, the test algorithm performs worse than the LSHADESPA algorithm, as indicated by the “−” sign. The final *t* circumstance happens when there is no statistical difference between the algorithms that are under test and is represented by the “=” symbol. The bold values in the table corresponds to the best results for that particular test problem. So, from the *w/l/t* row presented in the third last row of Table 5 and Fig. 9, it can be observed that the proposed LSHADESPA algorithm is significantly good for most of the problems in comparison with MH other algorithms. Thus, from the twelve problems, the EA4eig as well as NL-SHADE-RSP-MID is found to be best for four problem functions,

Table 3 Statistical outcomes of LSHADESPA algorithm for CEC 2021 ($D = 10$)

		Best	Worst	Median	Mean	Std
Basic	p_{rt1}	0	0	0	0	0
	p_{rt2}	0	4.93E−01	1.21E−01	1.32E−01	9.69E−02
	p_{rt3}	1.062E+01	1.042E+01	1.042E+01	1.052E+01	1.59E−15
	p_{rt4}	2.43E−01	4.87E−01	3.64E−01	3.60E−01	6.21E−02
	p_{rt5}	0	0	0	0	0
	p_{rt6}	9.64E−05	2.97E−01	2.38E−02	1.03E−01	1.42E−01
	p_{rt7}	1.24E−08	4.93E−01	7.03E−05	4.03E−02	1.26E−01
	p_{rt8}	0	0	0	0	0
	p_{rt9}	0	0	0	0	0
	p_{rt10}	4.762E+01	4.742E+01	4.772E+01	4.772E+01	1.09E−02
Bias	p_{rt1}	0	0	0	0	0
	p_{rt2}	0	3.11E−01	1.21E−01	1.47E−01	8.10E−02
	p_{rt3}	1.0612E+01	1.002E+01	1.0612E+01	1.0612E+01	1.062E+01
	p_{rt4}	2.36E−01	4.93E−01	3.35E−01	3.46E−01	7.04E−02
	p_{rt5}	0	2.06E−01	0	1.37E−02	5.26E−02
	p_{rt6}	1.70E−05	4.97E−01	1.97E−02	1.06E−01	1.64E−01
	p_{rt7}	2.48E−07	6.68E−02	3.63E−05	3.55E−04	1.18E−03
	p_{rt8}	0	0	0	0	0
	p_{rt9}	0	0	0	0	0
	p_{rt10}	4.772E+01	4.772E+01	4.772E+01	4.772E+01	1.08E−03
Shift	p_{rt1}	0	0	0	0	0
	p_{rt2}	6.23E−02	4.97E−01	1.85E−01	2.10E−01	1.07E−01
	p_{rt3}	1.062E+01	1.062E+01	1.062E+01	1.062E+01	1.52E−15
	p_{rt4}	2.04E−01	5.45E−01	3.64E−01	3.55E−01	7.92E−02
	p_{rt5}	0	1.162E+00	1.02E−01	4.762E+00	2.142E+01
	p_{rt6}	4.71E−04	3.382E+00	3.15E−02	3.55E−01	3.00E−01
	p_{rt7}	1.24E−05	1.162E+02	3.40E−04	7.952E+00	2.982E+00
	p_{rt8}	1.002E+02	1.002E+02	1.002E+01	1.002E+02	2.61E−15
	p_{rt9}	1.002E+02	3.252E+02	3.252E+01	3.072E+02	5.732E+01
	p_{rt10}	3.982E+02	3.981E+02	3.981E+02	3.981E+02	2.09E−04
Rotation	p_{rt1}	0	0	0	0	0
	p_{rt2}	2.47E−01	1.671E+01	3.641E+00	4.101E+00	4.291E+00
	p_{rt3}	1.031E+01	1.281E+01	1.131E+01	1.141E+01	6.85E−01
	p_{rt4}	3.00E−01	5.67E−01	4.04E−01	4.15E−01	7.43E−02
	p_{rt5}	0	1.181E+00	2.06E−01	2.64E−01	3.36E−01
	p_{rt6}	7.88E−03	1.171E+00	7.88E−04	2.52E−02	8.68E−02
	p_{rt7}	2.04E−05	3.78E−01	7.88E−04	2.52E−02	8.68E−02
	p_{rt8}	0	0	0	0	0
	p_{rt9}	0	0	0	0	0
	p_{rt10}	5.101E+01	7.151E+01	7.081E+01	6.751E+01	7.461E+00

Table 3 (continued)

		Best	Worst	Median	Mean	Std
Bias & shift	<i>p_{rt1}</i>	0	0	0	0	0
	<i>p_{rt2}</i>	0	3.10E−01	1.22E−01	1.50E−01	7.96E−02
	<i>p_{rt3}</i>	1.061E+01	1.061E+01	1.061E+01	1.061E+01	5.71E−14
	<i>p_{rt4}</i>	2.11E−01	5.25E−01	3.62E−01	3.58E−01	8.96E−02
	<i>p_{rt5}</i>	0	1.161E+02	0	8.311E+00	3.001E+01
	<i>p_{rt6}</i>	5.11E−06	4.95E−01	2.05E−02	1.03E−01	1.45E−02
	<i>p_{rt7}</i>	3.58E−07	4.91E−01	1.58E−03	7.11E−02	1.41E−01
	<i>p_{rt8}</i>	1.001E+02	1.001E+02	1.001E+02	1.001E+02	0
	<i>p_{rt9}</i>	3.192E+02	3.252E+02	3.252E+01	3.252E+02	1.192E+00
	<i>p_{rt10}</i>	4.002E+02	4.002E+02	4.002E+02	4.002E+02	0
Bias & rotation	<i>p_{rt1}</i>	0	0	0	0	0
	<i>p_{rt2}</i>	1.85E−01	7.012E+00	3.552E+00	2.212E+00	1.882E+00
	<i>p_{rt3}</i>	1.032E+01	1.242E+01	1.131E+01	1.132E+01	6.00E−01
	<i>p_{rt4}</i>	1.94E−01	4.71E−01	3.55E−01	3.65E−01	7.43E−02
	<i>p_{rt5}</i>	0	9.93E−01	2.06E−01	2.39E−01	2.25E−01
	<i>p_{rt6}</i>	1.10E−01	1.342E+00	5.19E−01	5.87E−01	3.18E−01
	<i>p_{rt7}</i>	2.10E−05	4.97E−01	2.00E−03	1.16E−01	1.79E−01
	<i>p_{rt8}</i>	0	0	0	0	0
	<i>p_{rt9}</i>	0	0	0	0	0
	<i>p_{rt10}</i>	5.102E+01	7.102E+01	7.102E+01	6.682E+01	8.012E+00
Shift & rotation	<i>p_{rt1}</i>	0	0	0	0	0
	<i>p_{rt2}</i>	3.10E−01	3.642E+01	5.352E+00	7.962E+00	8.092E+00
	<i>p_{rt3}</i>	1.012E+01	1.252E+01	1.132E+01	1.122E+01	5.71E−01
	<i>p_{rt4}</i>	2.22E−01	7.28E−01	3.96E−01	4.10E−01	1.06E−01
	<i>p_{rt5}</i>	2.06E−01	1.552E+02	4.14E−01	1.302E+01	3.512E+01
	<i>p_{rt6}</i>	2.13E−02	1.012E+00	2.31E−01	2.75E−01	2.55E−01
	<i>p_{rt7}</i>	2.26E−05	6.62E−01	3.48E−03	1.07E−01	1.87E−01
	<i>p_{rt8}</i>	1.002E+02	1.002E+02	1.002E+02	1.002E+02	2.63E−15
	<i>p_{rt9}</i>	1.002E+02	3.302E+02	3.262E+01	3.112E+02	5.782E+01
	<i>p_{rt10}</i>	3.952E+02	4.432E+02	4.182E+02	4.182E+02	2.302E+01
Bias, shift rotation	<i>p_{rt1}</i>	0	0	0	0	0
	<i>p_{rt2}</i>	2.48E−01	2.833E+01	6.963E+01	9.443E+00	7.563E+00
	<i>p_{rt3}</i>	1.013E+01	1.253E+01	1.123E+01	1.123E+01	5.35E−01
	<i>p_{rt4}</i>	1.84E−01	5.57E−01	3.79E−01	3.79E−01	9.14E−02
	<i>p_{rt5}</i>	0	3.883E+01	4.14E−01	3.253E+00	7.693E+00
	<i>p_{rt6}</i>	2.68E−02	1.013E+00	2.44E−01	2.68E−01	2.04E−01
	<i>p_{rt7}</i>	4.56E−06	6.18E−01	1.08E−03	8.19E−02	1.64E−01
	<i>p_{rt8}</i>	1.003E+02	1.003E+02	1.001E+02	1.001E+02	0
	<i>p_{rt9}</i>	1.001E+02	3.301E+02	3.281E+02	3.111E+02	5.781E+01
	<i>p_{rt10}</i>	3.951E+02	4.411E+02	3.962E+02	4.152E+02	2.272E+01

and DE for two problems as compared to the proposed LSHADESPA algorithm. In addition to that, the Friedman rank test is also utilized to validate the superior performance of the proposed LSHADESPA algorithm. In this test, every algorithm provides a rank according to its performance, and also average f-rank is obtained from all the

Table 4 Statistical outcomes of LSHADESPA algorithm for CEC 2021 ($D = 20$)

		Best	Worst	Mean	Median	Std
Basic	$Fprt_1$	0	0	0	0	0
	p_{rt_2}	1.851E−01	1.902E+00	2.791E−01	3.362E+00	3.021E−01
	p_{rt_3}	2.002E+01	2.002E+01	2.002E+01	2.002E+01	1.061E−14
	p_{rt_4}	5.641E−01	1.301E+00	1.061E+00	1.031E+00	1.961E−01
	p_{rt_5}	0	2.061E−01	0	3.451E−02	6.861E−02
	p_{rt_6}	4.711E−01	6.481E−01	5.261E−01	5.281E−01	4.301E−02
	p_{rt_7}	6.431E−02	5.001E−01	4.971E−01	4.821E−E−01	7.911E−02
	p_{rt_8}	0	0	0	0	0
	p_{rt_9}	0	0	0	0	0
	$p_{rt_{10}}$	4.851E+01	4.861E+01	4.851E+01	4.851E+01	2.001E−02
Bias	p_{rt_1}	0	0	0	0	0
	p_{rt_2}	1.541E−01	1.841E+00	2.791E−01	3.201E−01	2.961E−01
	p_{rt_3}	2.001E+01	2.001E+01	2.001E+01	2.001E+01	2.051E−14
	p_{rt_4}	7.462E−01	1.441E+00	1.011E+00	1.011E+00	1.712E−01
	p_{rt_5}	0	2.062E−01	0	1.712E−02	4.782E−02
	p_{rt_6}	4.912E−01	6.38−01	5.312E−01	5.352E−01	3.521E−02
	p_{rt_7}	8.861E−02	6.541E−01	4.961E−01	4.681E−01	9.951E−02
	p_{rt_8}	0	0	0	0	0
	p_{rt_9}	0	0	0	0	0
	$p_{rt_{10}}$	4.864E+01	4.864E+01	4.864E+01	4.854E+01	1.961E−02
Shift	p_{rt_1}	0	1.004E+00	0	0	0
	p_{rt_2}	1.851E−01	7.784E+00	3.411E−01	1.164E+00	1.784E+00
	p_{rt_3}	2.004E+01	2.004E+01	2.004E+01	2.004E+01	1.013E−14
	p_{rt_4}	7.603E−01	1.234E+00	9.933E−E−01	1.004E+00	1.443E−01
	p_{rt_5}	0	1.164E+02	1.023E−01	1.284E+01	3.573E+01
	p_{rt_6}	4.502E−01	6.002E−01	4.972E−01	5.062E−01	2.842E−02
	p_{rt_7}	5.002E−03	1.163E+02	4.971E−01	1.593E+01	4.073E+01
	p_{rt_8}	1.003E+02	1.003E+02	1.003E+02	1.003E+02	0
	p_{rt_9}	3.923E+02	4.003E+02	3.973E+02	3.972E+02	1.482E+00
	$p_{rt_{10}}$	4.792E+02	4.832E+02	4.822E+02	4.822E+02	1.472E+00
Rotation	p_{rt_1}	0	0	0	0	0
	p_{rt_2}	2.471E−01	9.371E+00	2.061E+00	2.791E+00	2.151E+00
	p_{rt_3}	2.051E+01	2.421E+01	3.341E+01	2.231E+01	9.641E−01
	p_{rt_4}	8.161E−01	1.461E+00	1.061E+00	1.091E+00	1.911E−01
	p_{rt_5}	1.592E+00	1.132E+01	4.832E+00	5.232E+00	2.412E+00
	p_{rt_6}	4.763E−01	3.382E+00	1.222E+00	1.492E+00	1.042E+00
	p_{rt_7}	7.493E−01	3.522E+00	1.602E+00	1.762E+00	7.302E−01
	p_{rt_8}	0	0	0	0	0
	p_{rt_9}	0	0	0	0	0
	$p_{rt_{10}}$	6.212E+01	6.232E+01	6.222E+01	6.222E+01	3.222E−02

Table 4 (continued)

		Best	Worst	Mean	Median	Std
Bias & shift	<i>p_{rt1}</i>	0	0	0	0	0
	<i>p_{rt2}</i>	1.541E−01	3.582E+00	3.572E+00	8.421E−01	9.012E−01
	<i>p_{rt3}</i>	2.001E+01	2.001E+01	2.001E+01	2.001E+01	3.442E−14
	<i>p_{rt4}</i>	6.061E−01	1.471E+00	1.041E+00	1.041E+00	2.211E−01
	<i>p_{rt5}</i>	0	1.171E+02	1.022E−01	8.811E+00	3.001E+01
	<i>p_{rt6}</i>	2.322E−01	6.021E−01	4.991E−01	4.993E−01	5.893E−02
	<i>p_{rt7}</i>	1.093E−01	1.161E+02	4.973E−01	2.781E+01	5.071E+01
	<i>p_{rt8}</i>	1.001E+02	1.001E+02	1.001E+02	1.001E+02	8.282E−14
	<i>p_{rt9}</i>	3.921E+02	4.011E+02	4.001E+02	4.001E+02	1.303E+00
	<i>p_{rt10}</i>	4.793E+02	4.833E+02	4.833E+02	4.823E+02	1.243E+00
Bias & rotation	<i>p_{rt1}</i>	0	0	0	0	0
	<i>p_{rt2}</i>	2.162E−01	9.433E+00	2.033E+00	3.143E+00	2.401E+00
	<i>p_{rt3}</i>	2.011E+01	2.511E+01	2.251E+01	2.262E+01	1.172E+00
	<i>p_{rt4}</i>	6.662E−01	1.482E+00	1.022E+00	1.021E+00	1.832E−01
	<i>p_{rt5}</i>	1.491E+00	9.201E+00	4.281E+00	4.401E+00	1.701E+00
	<i>p_{rt6}</i>	4.762E−01	3.741E+00	1.011E+00	1.393E+00	1.003E+00
	<i>p_{rt7}</i>	7.612E−01	3.163E+00	1.553E+00	1.633E+00	6.801E−01
	<i>p_{rt8}</i>	0	0	0	0	0
	<i>p_{rt9}</i>	0	0	0	0	0
	<i>p_{rt10}</i>	6.213E+01	6.213E+01	6.223E+01	6.223E+01	2.301E−02
Shift & rotation	<i>p_{rt1}</i>	0	0	0	0	0
	<i>p_{rt2}</i>	3.431E−01	3.403E+02	1.043E+01	3.602E+01	7.512E+01
	<i>p_{rt3}</i>	2.022E+01	2.472E+01	2.232E+01	2.242E+01	1.082E+00
	<i>p_{rt4}</i>	6.071E−01	1.332E+00	1.052E+00	1.022E+00	2.06−01
	<i>p_{rt5}</i>	1.112E+01	1.512E+02	3.252E+01	6.551E+01	5.551E+01
	<i>p_{rt6}</i>	5.151E−01	2.301E+00	1.491E+00	1.171E+00	6.071v01
	<i>p_{rt7}</i>	1.921E+00	1.241E+02	4.921E+00	1.281E+01	3.001E+01
	<i>p_{rt8}</i>	1.001E+02	1.001E+02	1.001E+02	1.001E+02	0
	<i>p_{rt9}</i>	3.962E+02	4.012E+02	4.012E+02	4.012E+02	1.412E+00
	<i>p_{rt10}</i>	4.102E+02	4.112E+02	4.112E+02	4.112E+02	6.162E−01
Bias, shift, rotation	<i>p_{rt1}</i>	1.002E+10	1.002E+10	1.002E+10	1.002E+10	0
	<i>p_{rt2}</i>	1.813E+01	2.303E+02	6.913E+00	2.193E+01	4.973E+01
	<i>p_{rt3}</i>	2.043E+01	2.453E+01	2.233E+01	2.233E+01	1.053E+00
	<i>p_{rt4}</i>	7.17512E−01	1.533E+00	1.063E+00	1.083E+00	1.9612E−01
	<i>p_{rt5}</i>	1.573E+01	2.503E+02	5.403E+01	8.603E+01	6.403E+01
	<i>p_{rt6}</i>	5.3812E−01	2.041E+00	1.491E+00	1.171E+00	5.2011E−01
	<i>p_{rt7}</i>	1.611E+00	1.211E+01	4.521E+00	4.521E+00	2.001E+00
	<i>p_{rt8}</i>	1.001E+02	1.004E+02	1.004E+02	1.004E+02	1.121E−13
	<i>p_{rt9}</i>	3.994E+02	4.034E+02	4.014E+02	4.014E+02	1.051E+00
	<i>p_{rt10}</i>	4.101E+02	4.091E+02	4.111E+02	4.111E+02	8.171E−01

problem displayed in Table 5. From Table 5 and Fig. 10, it can be observed that the LSHADESPA algorithm obtained 1st rank followed by NL-SHADE-RSP-MID and the EA4eig algorithm.

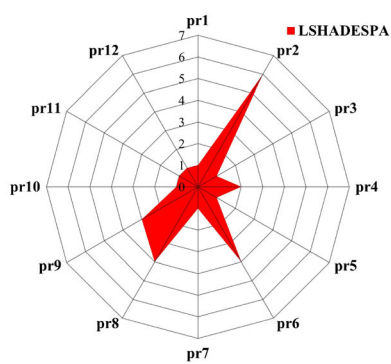
The LSHADESPA algorithm is the most effective, achieving the highest rank in 41.67% of test instances and exhibiting an efficiency of 81.95% compared to the optimal ranking. The NL-SHADE-RSP-MID follows closely, winning 33.33% of the cases with a strong 76.39% average rank efficiency. DE and PSO also achieve a 33.33% win rate, but their overall ranking efficiency is lower, with DE at 58.33% and PSO at 41.67%, indicating

Table 5 Outcomes of LSHADESPA algorithm in comparison to other MH algorithms for CEC 2022 problems

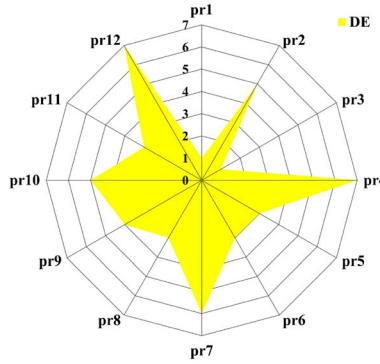
Problem		LSHADESPA	DE [58]	PSO [58]	LSO [59]	EA4eig [51]	NL-SHADE-RSP-MID [61]	KOA [48]
pr_1	Mean	3.000E+02	3.000E+02	3.000E+02	3.000E+02	3.000E+02	3.001E+02	3.00E+02
	Std	0.000E+00	0.000E+00	2.585E−16	0.000E+00	1.340E−09	0.000E+00	1.06E−14
	p-rank		=	—	=	—	=	—
	f-rank	1	1	5	1	7	1	6
pr_2	Mean	4.062E+02	4.070E+02	4.235E+02	4.009E+02	4.014E+02	4.000E+02	4.02E+02
	Std	2.2424E+00	2.090E+00	3.033E+01	2.13E+00	1.953E+00	0.000E+00	3.39E+00
	p-rank		—	—	—	—	+	—
	f-rank	6	5	7	2	3	1	4
pr_3	Mean	6.000E+02	6.000E+02	6.003E+02	6.000E+02	6.000E+02	6.000E+02	6.00E+02
	Std	0.000E+00	0.000E+00	6.140E−01	0.000E+00	9.980E−10	0.000E+00	6.33E−14
	p-rank		=	—	=	—	=	—
	f-rank	1	1	7	1	6	1	5
pr_4	Mean	8.015E+02	8.167E+02	8.114E+02	8.107E+02	8.012E+02	8.047E+02	8.044E+02
	Std	7.498E−01	2.956E+00	5.123E+00	2.96E+00	1.042E+00	1.330E+00	1.62 E+00
	p-rank		—	—	—	+	—	—
	f-rank	2	7	6	5	1	4	3
pr_5	Mean	9.000E+02	9.000E+02	9.001E+02	9.000E+02	9.000E+02	9.000E+02	9.000E+02
	Std	0.000E+00	1.055E−13	1.181E−01	8.29E−02	1.620E−09	0.000E+00	2.00E−02
	p-rank		—	—	—	—	=	—
	f-rank	1	3	7	6	4	1	5
pr_6	Mean	1.800E+03	1.800E+03	5.308E+03	1.800E+03	1.800E+03	1.800E+03	1.80E+03
	Std	2.522E−01	1.428E−01	2.651E+03	3.400E−01	3.570E−02	1.270E−01	1.80E+03
	p-rank		+	—	—	+	+	—
	f-rank	4	3	7	5	1	2	6
pr_7	Mean	2.000E+03	2.001E+03	2.013E+03	2.000E+03	2.000E+03	2.000E+03	2.000E+03
	Std	1.602E−11	3.668E+00	9.796E+00	5.400E−01	1.170E−E−09	0	1.100E−01
	p-rank		—	—	—	—	—	—
	f-rank	1	6	7	5	3	2	4
pr_8	Mean	2.201E+03	2.200E+03	2.227E+03	2.201E+03	2.200E+03	2.200E+03	2.20E+03
	Std	4.741E+00	3.865E−01	3.930E+01	3.83E+00	6.810E−02	2.730E−01	6.20E+00
	p-rank		+	—	—	+	+	—
	f-rank	4	3	7	5	2	1	6
pr_9	Mean	2.525E+03	2.529E+03	2.534E+03	2.529E+03	2.485E+03	2.491E+03	2.529E+03
	Std	3.214E+00	0.000E+00	1.202E+01	5.680v14	8.540E+01	6.350E−02	0.000E+00
	p-rank		—	—	—	+	+	—
	f-rank	3	4	7	6	1	2	6
pr_{10}	Mean	2.500E+03	2.542E+03	2.572E+03	2.500E+03	2.500E+03	2.600E+03	2.500E+03
	Std	3.703E−02	4.934E+01	9.683E+01	6.000E−01	3.600E−02	0.000E+00	4.000E−01
	p-rank		—	—	—	—	—	—
	f-rank	1	5	6	4	2	7	3
pr_{11}	Mean	2.600E+03	2.660E+03	2.763E+03	2.825E+03	2.600E+03	2.862E+03	2.770E+03
	Std	0.000E+00	8.529E+01	1.178E+02	1.291E+02	1.060E−09	1.770E+00	151.20
	p-rank		—	—	—	—	—	—
	f-rank	1	3	4	6	2	7	5

Table 5 (continued)

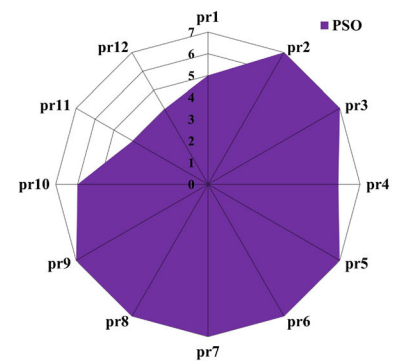
Problem		LSHADESPA	DE [58]	PSO [58]	LSO [59]	EA4eig [51]	NL-SHADE-RSP-MID [61]	KOA [48]
pr_{12}	Mean	2.862E+03	2.954E+03	2.871E+03	2.862E+03	2.847E+03	2.865E+03	2.864E+03
	Std	2.970E+00	6.279E-02	1.292E+01	1.72E+00	3.901E+00	9.720E-01	1.16E+00
	p-rank		—	—	=	—	—	—
	f-rank	1	7	4	1	6	5	3
w/l/t		NA	2/8/2	0/12/0	0/9/3	4/8/0	4/5/3	0/12/0
Average f-rank		2.166	4.000	6.166	3.916	3.167	2.833	4.667
Overall f-rank		1	5	7	4	3	2	6



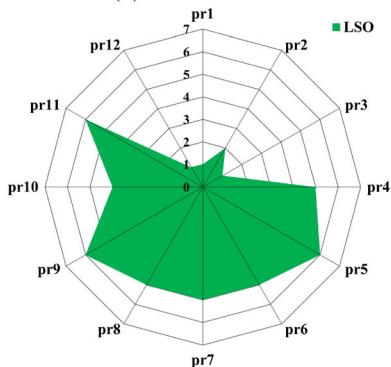
(a) LSHADESPA



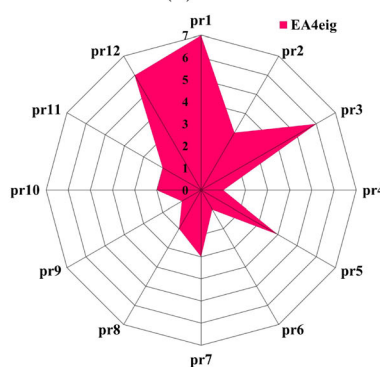
(b) DE



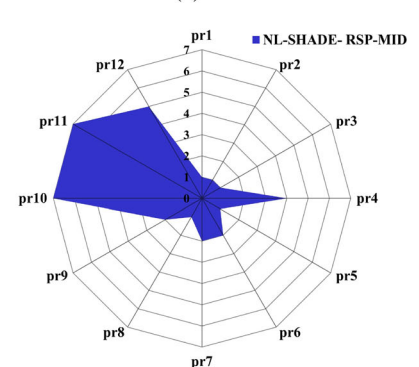
(c) PSO



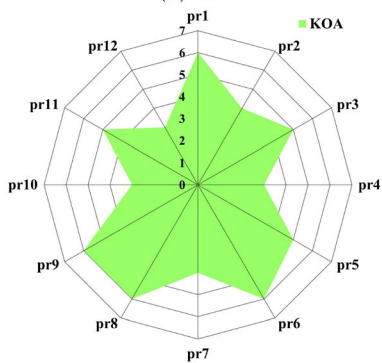
(d) LSO



(e) EA4eig



(f) NL-SHADE- RSP-MID



(g) KOA

Fig. 8 Radarchart of LSHADESPA, DE, PSO, LSO, EA4eig, MVMO, NL-SHADE- RSP-MID, and KOA for CEC 2022 numerical challenges

Fig. 9 Comparison of the performance of the LSHA-DESPA algorithm with other algorithms for CEC 2022 (w/l/t scenario)

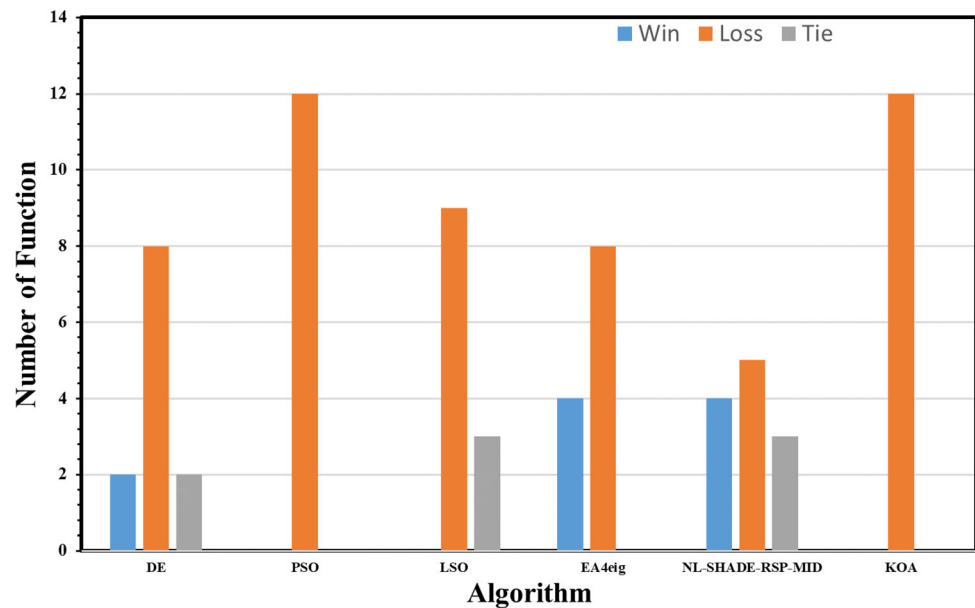
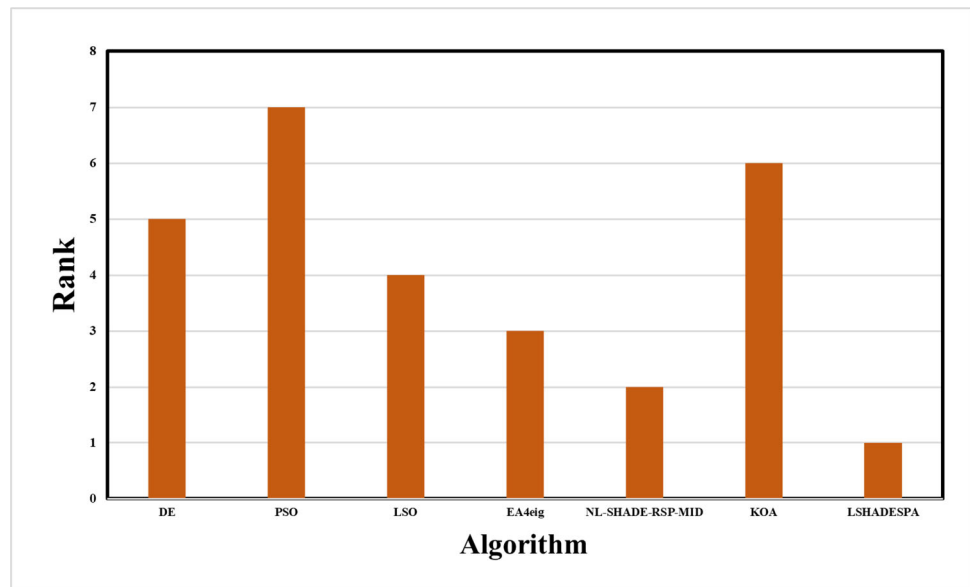


Fig. 10 Rankings of algorithms for CEC 2022



inconsistency in performance. LSO maintains a moderate 25% win rate with a 67.37% ranking efficiency, while EA4eig and KOA lag behind, winning only 16.67% of the cases, with ranking efficiencies of 73.61% and 61.11%, respectively. Overall, LSHADESPA proves to be the most stable and high-performing algorithm, whereas PSO is the weakest, showing the lowest ranking efficiency.

5 Discussion

This section is structured into three subsections. The first subsection provides an overall summary of the work. The second subsection focuses on the drawback of the LSHADESPA algorithm. Lastly, the third subsection discusses potential future Insightful Implications. The details are presented below, subsection.

5.1 Summary of results

The overall summary of the presented outcomes is given below:

- In this paper, we introduced a new variant of the DE algorithm known as the LSHADESPA algorithm. By incorporating the SP mechanism, SA scaling factor, and oscillating inertia weight for added exploration and exploitation properties.
- In the CEC 2014 benchmark challenges, which included uni-modal, multi-modal, and composition challenges, the performance of the LSHADESPA algorithm outperformed that of the IMEHO, LX-BBO, PBIL, B-BBO, VNBA, RW-GWO, ISOS, as well as CCS algorithms.
- The proposed LSHADESPA algorithm has been evaluated using the CEC 2017 benchmark dataset. The evaluation outcomes indicate that the LSHADESPA algorithm is superior to other MH algorithms such as SaDE, JADE, SHADE, MVMO, CV1.0, CVnew, and SACS.
- The outcomes obtained from the simulation for CEC 2021 are also presented in this paper. The LSHADESPA algorithm provides outcomes that closely match the optimal solutions for the CEC 2021 test suite.
- Additionally, the LSHADESPA algorithm has superior performance compared to the PSO, LSO, DE, EA4eig, NL-SHADE-RSP-MID, and KOA algorithms, as observed in the comparison with CEC 2022 benchmarks.
- The statistical significance of the LSHADESPA algorithm has been evaluated utilizing Friedman and Wilcoxon's tests. These tests provide valuable insights into the algorithm's performance and potential impact.
- Overall, these results prove that the added enhancements make the algorithm perform better in terms of exploration and exploitation and hence result in an improved search operation.

5.2 Drawbacks

- Despite LSHADESPA's high performance, it is not guaranteed to find global optimum solutions for all domain research problems due to its stochastic nature.
- Due to the resizing of population and mutation characteristics, there is a possibility that the computational complexity becomes $O(\log N)$. Although it becomes occasionally and is not a significant factor, but when the problem dimensions increase, the complexity also grows, and this component can become more significant. Thus, the complexity can play a major role and hence can be considered as a drawback for higher-dimensional analysis.
- In addition, LSHADESPA's poor performance is shown in comparison outcomes on benchmark challenges. The algorithm performed well on CEC 2014, CEC 2017, CEC 2021, and 2022, where the global optima were quite different from the actual values obtained through the algorithm for some functions. It means that more work is needed to develop an ideal algorithm that solves all optimization problems.

5.3 Insightful implications

- The LSHADE algorithm can be included in the list of expert as well as hybrid intelligent systems due to its simplicity and linear design.
- In order to optimize algorithms for a wide range of real-world problems, it is essential to incorporate an accurately calibrated equilibrium between exploration and exploitation.
- Further investigation can be done to examine the algorithm's effectiveness by comprehensively examining its stability, convergence features, and other empirical studies. In addition to this, it is also possible to conduct theoretical investigations to enhance knowledge of the indicated LSHADE algorithm.

- The majority of research problems in the domain demonstrate a multi-objective nature. Therefore, it is advantageous to explore algorithms that possess enhanced features, as they can be extended to accommodate the multi-objective nature of these problems.

6 Conclusion

This study offers a new variant of LSHADE denoted by LSHADESPA algorithm that has been applied to CEC 2014, CEC 2017, CEC 2021 as well as CEC 2022 benchmark problems. The proposed algorithm is improved by incorporating self-adaptive features such as *SP* adaptation to minimize computational burden, *SA* modification to adjust scaling factors to achieve superior exploration characteristics, as well as oscillating adaptation-oriented crossover rate to find a balanced exploration and exploitation operation. All these added properties are tested using four benchmark datasets, namely CEC 2014, CEC 2017, CEC 2021, and CEC 2022. The algorithm is compared to SaDE, JADE, LSHADE, SHADE, CmA-ES, MVMO, SACS, VNBA, and others, and it has been found that LSHADESPA performs better. Further, Wilcoxon rank sum and Friedman test shows that the algorithm is also better significantly.

In the future, the LSHADESPA algorithm can be used for image thresholding, genome sequencing, clustering, and other real-world applications. The algorithm can also be enhanced for better exploration and exploitation properties for improved performance. Because of the better search properties of LSHADESPA, a binary and constraint version can be introduced for discrete and other optimization problems.

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Data availability The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare that there is no conflict of interest associated.

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